



STUDY PROTOCOL

REVIEWED Ambient and Indoor Air Pollution in Pregnancy and the risk of Low birth weight and Ensuing Effects in Infants (APPLE): A cohort study in Bangalore, South India

[version 4; peer review: 6 approved with reservations, 1 not approved]

Prafulla Shriyan ^{1*}, Giridhara R Babu ^{1*}, Deepa Ravi ¹, Yamuna Ana ¹, Onno CP van Schayck², Prashanth Thankachan³, GVS Murthy ⁴

¹Indian Institute of Public Health, Public Health Foundation of India, Bangalore, Karnataka, 560023, India

²Care and Public Health Research Institute, Maastricht University, Maastricht, Limburg, The Netherlands

³Division of Nutrition, St Johns Research Institute Bangalore, Bangalore, Karnataka, 560034, India

⁴International Centre for Eye Health, Department of Clinical Research, London School of Hygiene & Tropical Medicine, London, WC1E 7HT, UK

* Equal contributors

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Abstract

Background

Exposure to air pollution (IAP) from the combustion of solid fuels is a significant cause of morbidity and mortality in developing countries. Pregnant women exposed to higher pollutant levels are at higher risk of delivering a low-birth-weight (LBW) baby. There is a lack of standardized data regarding the levels and types of specific pollutants and how they impact LBW. We aim to prospectively assess the association between ambient and indoor air pollution levels in pregnancy and low birth weight and understand the subsequent risk of adiposity in these infants.

Methods

We will conduct a prospective cohort study of 516 pregnant women recruited before 18 weeks of gestation in the urban slums of Bangalore, who have voluntarily consented to participate. We will

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1. Kalpana Balakrishnan , Sri Ramachandra Institute of Higher Education and Research, Chennai, India
2. Ryan W. Allen, Simon Fraser University, Burnaby, Canada

estimate the level of air pollutants including particulate matter ($\leq 10 \mu\text{m}$, $\leq 2.5 \mu\text{m}$) and carbon monoxide (CO) parts per million (ppm) levels in both indoor and ambient environment. The follow-up of the delivered children will be done at delivery until the infant is one year old. The association between pollutants and LBW will be evaluated using logistic regression adjusting for potential confounders. Further, we will explore the mediation role of LBW in the hypothesized causal chain of air pollution and adiposity. Nested within a large cohort titled “Maternal Antecedents of Adiposity and Studying the Transgenerational role of Hyperglycemia and Insulin (MAASTHI)”, we can estimate the absolute risk of having low birth weight caused by air pollution and other variables.

Discussion

Understanding the association between exposures to ambient and indoor air pollution and low birth weight is essential in India. LBW babies have a higher risk of developing obesity and Non-Communicable Diseases (NCDs) during adulthood. The results from this study can inform the efforts for controlling the air pollution-related chronic diseases in India.

Keywords

Ambient air pollution, indoor air pollution, low birth weight, adiposity, Non-communicable disease



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3. **Julian D. Marshall**, University of Washington, Seattle, USA

4. **Ebba Malmqvist**, Lund University, Lund, Sweden

5. **Chan Lu**, Central South University, Changsha, China

6. **Jiawen Liao**, University of Southern California, Los Angeles, USA

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Corresponding author: Giridhara R Babu (epigiridhar@gmail.com)

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REVISED Amendments from Version 3**Exposure assessment**

New subsection describing how two 24-hour personal measurements are temporally adjusted against the continuous ambient record to derive trimester-specific and pregnancy-average exposures, with ICC testing, regression calibration and an explicit statement of the expected direction of bias.

Sensor calibration

New subsection on calibration against a co-located E-BAM reference instrument ($R^2 > 0.88$), device-specific correction equations, humidity correction for hygroscopic growth, and periodic re-checks. The broken manufacturer link has been removed. Calibration scatter plots added as Supplementary Figures 1 and 2.

Confounder set

Traffic and roadway proximity removed from the confounder set, with the over-adjustment rationale stated explicitly. Traffic retained as a determinant of exposure and a candidate effect modifier.

Statistical power

Post-hoc power replaced with a minimum detectable effect analysis reported against defined exposure contrasts, together with an explicit statement about the exposure gradient.

Housing and cooking covariates

Expanded description of the ventilation, housing, cooking-location, fuel, cook identity, cooking duration and time-activity data already collected.

Adiposity mechanism

New Background paragraph setting out the placental, inflammatory, adipokine and catch-up-growth pathways linking prenatal air pollution to infant adiposity.

Analysis plan

Planned sensitivity analyses stratified by personal and household factors added, together with sensitivity analyses for exposure windows, measurement error, humidity correction and live-birth bias.

Laboratory section

Role of MEDALL under the memorandum of understanding clarified; timing of blood collection specified as at or after 24 weeks of gestation.

Figure 2

Map redrawn with a labelled lat/long graticule, north arrow, scale bar, legend, stated projection and base map attribution on both panels; panels labelled A and B; facility label spelling corrected.

Notation

PM10 ($\leq 10 \mu\text{m}$) and PM2.5 ($\leq 2.5 \mu\text{m}$) notation corrected throughout the abstract and manuscript.

References

Eleven references added (53–63). Existing numbering 1–52 is unchanged.

Discussion

Strengths, limitations and an environmental justice/future work paragraph added.

Any further responses from the reviewers can be found at the end of the article

List of Abbreviation

AQI: Air Quality Index

ATP: Adenosine TriPhosphate

BMI: Body Mass Index

CO: Carbon Monoxide

CSV: comma separated values

DNA: Deoxyribose Nucleic Acid

HBsAg: Hepatitis B Surface Antigen

HIV: Human Immunodeficiency Virus

IAP: Indoor Air Pollution

LBW: Low Birth Weight

NCD: Non-Communicable Disease

NO₂: Nitrogen Dioxide

PM10: Particulate matter 10 micrometre

PM2.5: Particulate matter 2.5 micrometer

PPM: parts per million
 RCT: Randomized Control Trial
 RH: Relative Humidity
 SD: Secure Digital
 SES: Socio-Economic Status
 SO₂: Sulphur Dioxide
 T2DM: Type 2 Diabetes Mellitus
 USB: Universal Serial Bus
 US EPA: The United States Environmental Protection Agency
 VDRL: Venereal Disease Research Diagnostic
 XLSX: Excel Microsoft Office Open XML Format Spreadsheet file

Background

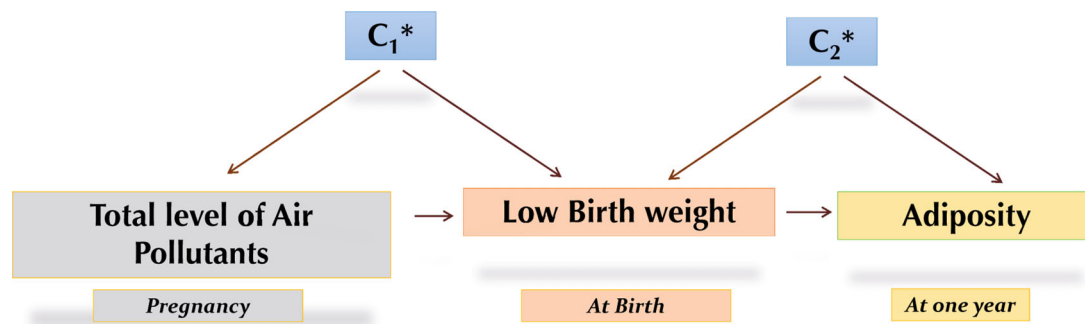
The developmental origins of health and disease (DOHaD) hypothesis deal with exploring the casual role of intrauterine circumstances to the origins of diseases in adults. The initial studies revealed that undernutrition during pregnancy is an important determinant of adult cardiac and metabolic disorders due to fetal programming. This was hypothesized to be mediated via altering the fetus' structure, function, and metabolism. Since then, development of fetal origins of adult disorders has remained an essential focus of researchers in the exploration of causal mechanisms for hypertension, coronary heart disease, and non-insulin-dependent diabetes. The collation of the prospective evidence using the DOHaD approach helps in assessing and modifying the impact of determinants of health from the life-course perspective. While the evidence regarding nutritional pathways of LBW is available, other antecedents such as air pollution and psychosocial stress are less investigated.¹

Globally, nearly three billion people use traditional biomass fuels as their primary source of energy comprising of wood, charcoal and agricultural wastes.^{2,3} In India, nearly 67% of the population use biomass fuel as the primary source for cooking. As a result, exposure to indoor air pollution (IAP) from the combustion of these fuels has emerged as an important cause of morbidity and mortality.^{4,5} Air pollution is contributing to the second-highest associated risk factor for mortality and morbidity.⁶ Even a low-dose exposure to pollutants *in utero* can result in disease, disability and death in childhood and across their lifespan.³ There is evidence that air pollution is associated with the burden of several diseases including respiratory infections, chronic obstructive pulmonary disease, cataracts, cardiovascular events, low birth weight and all-cause mortality. A meta-analysis indicates that the decrease in birth weight is proportional to higher pollutant concentration.⁷

The putative role of exposure to air pollution during pregnancy resulting in LBW has been assessed in several studies.^{1,7-13} The suggested mechanisms mediating this path include oxidative stress resulting in placental and endothelial dysfunction,¹⁴ and damage in the DNA productivity due to an imbalance between reactive oxygen species.¹⁵ Specifically, exposure to particulate matter during pregnancy induces changes in multiple placental compartments, including the maternal vascular space, fetal capillaries, and surface exchange areas.¹⁶ These alterations in placental function were associated with a higher incidence of LBW among exposed fetuses.^{16,17} The poor nutrition accentuates the propensity of a baby to be LBW and subsequent inadequate development of pancreatic beta cell mass resulting in a higher risk of development of type 2 diabetes in future. These include greater insulin resistance and storage of fat as compared to children with normal weight. Intrauterine malnutrition and other fetal constraints induce insulin deficiency (lack of the growth-promoting activities of insulin) and a postnatal state of regulatory insulin resistance, which leads to a rapid postnatal increase of adipose tissue that remains stable throughout life.¹⁸ Maternal exposure to air pollutants induces placental dysfunction and systemic inflammation, which disrupts the fetal energy homeostasis and adipokine regulation.^{1,2} These triggers impair insulin signaling and promote lipidomic changes that 'program' the fetus for altered adipose tissue development, significantly increasing long-term obesity risks.^{3,4}

Available evidence suggests that several environmental factors induce intra uterine growth retardation (IUGR) and subsequent LBW in newborns. These include diet,¹⁹ diabetes,²⁰ hormone exposure,²¹ air pollution,^{5,6,22} psychosocial stressors²³ and hypoxia.^{24,25} Our ongoing, MAASHTI birth cohort study characterizes the impact of these factors, including exposure to higher glucose levels during pregnancy, in the intrauterine milieu on the fetus, barring the effect of air pollution. A cohort study is an efficient study design to assess the association between prenatal exposure to air pollutants and infant health outcomes.

Objectives: We aim to evaluate the association between prenatal exposure to total level of air pollutants and low birth weight in newborns (Figure 1). We will also be exploring the mediation role played by LBW in the causal path between air pollution in pregnancy and adiposity in infancy (Figure 1).



***Potential confounders**

- C1: Household/indoor temperature, Season, Maternal smoking(active and passive), Socio-economic status, Antenatal stress, Maternal Age, Occupational H/o exposure, Ambient temperature, Heavy traffic roadways
- C2: SES, Gender, Feeding habits, Physical activity, Exact age at measurement, Feeding habits

Figure 1. Directed acyclic graph displaying exposure, intermediate and outcome of interest. *Potential confounders. C1: Household/indoor temperature, season, maternal smoking (active and passive), socioeconomic status, antenatal depression, maternal age, occupational H/O exposure, and ambient temperature. C2: Child sex, feeding habits, and child age.

1. To explore the association of prenatal exposure to total level of air pollutants in pregnancy and low birth weight at birth.

Hypothesis 1: Prenatal exposure to higher levels of air pollution increases the risk of low birth weight.

2. To evaluate the association between prenatal exposure to total level of air pollutants in pregnancy and adiposity in infants, mediated through low birth weight.

Hypothesis 2: Low birth weight at birth mediates the effect of air pollution in pregnancy on adiposity in children.

Methods

Study design

A prospective cohort study is planned in the urban slums of Bangalore. The study duration is three years. In this regard, a pilot study has been started in September 2017, following which the recruitment will begin, and follow-ups will continue up to 2020. The study population comprises of households with pregnant women. All the pregnant women in the study population will be followed up until delivery, and their infants will be followed up further irrespective of the exposure status. The follow-ups will be performed at delivery and one year. Infant anthropometry, morbidity, feeding practices, and child developmental milestones will be assessed during each follow-up visit (Figure 3).

Setting: The study area is located in the slums of east and west zones of urban Bangalore. The selected areas are Srirampura, Kodandarampura, Shirdi Saiba Nagar, Subasnagar from the West Zone, and DJ Halli, Bagalur Layout and Pulikeshinagar from the East Zone. The recruitment will be initiated in health facilities like Srirampura Referral Hospital and New Bagalur Layout Urban Health Centres, serving to these areas (Figure 2). A slum is defined as an area comprising of at least 60–70 households living in poorly built congested tenements along with the neighbouring well-built houses (<http://ksdb.kar.nic.in/slums.asp>).

Participants: The study participants will be selected from the health centres. Permissions have been obtained from the Chief Health Officer, Bangalore to conduct the study in the selected BBMP urban health centres. The research staff will screen the eligible participants in the health centre and in the community. Only eligible participants will be enrolled in the study after obtaining their informed consent.

Eligibility criteria: Pregnant women aged between 18–45 years with a gestational age of under 18 weeks who reside in the slums and plan to deliver at the study locations are eligible for recruitment in the study. Women with severe co-existing illness and those who plan to move out of the study location during the study period will be excluded. Women who are mentally or physically not capable of voluntarily consenting or participating in the study will be excluded.

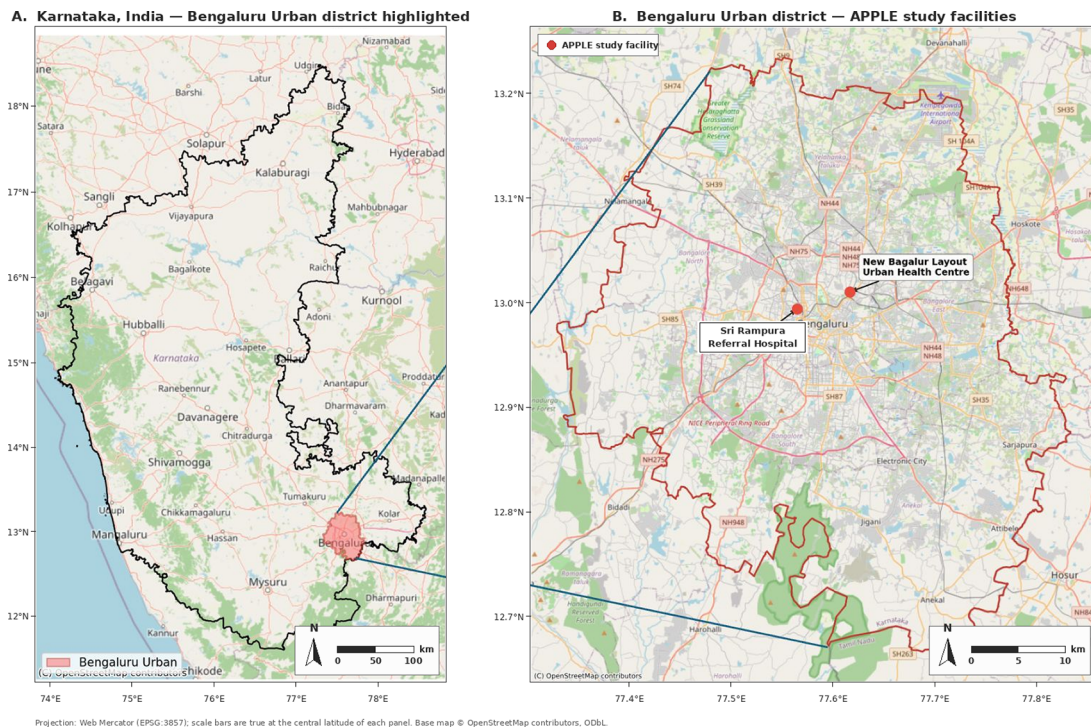


Figure 2. Map of the study site. The red marker indicates the health facilities in Bengaluru Urban area in Karnataka, India, where the APPLE cohort study was conducted.

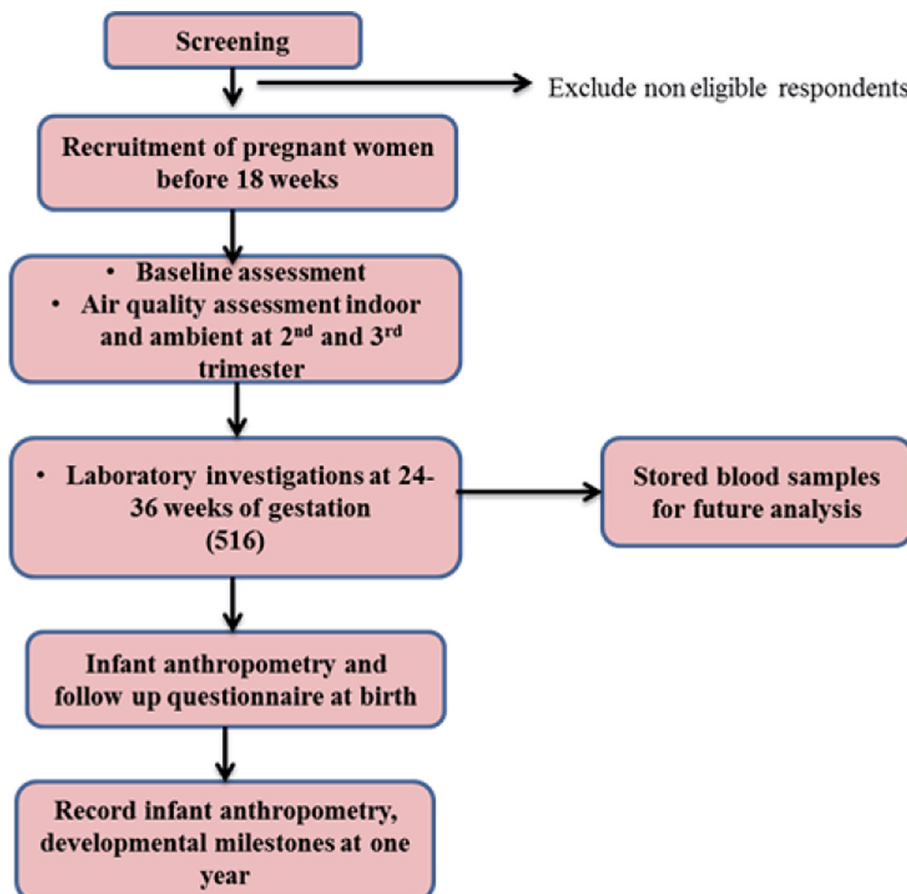


Figure 3. Flow diagram depicting the steps in a cohort study.

Ethical considerations

Institutional Ethics Committee: Ethical clearance for the proposed study has been obtained from the institutional review board (IEC) at Bangalore, IIPH-H (Approval Number IIPHHB/TRCIEC/121/2017). Written informed consent will be obtained from participants before the start of the study. They will be informed in detail about the study particulars and their voluntary participation before obtaining consent.

Variables

Exposure assessment: We will assess the total level of air pollutants namely fine particulate matter ($\leq 2.5 \mu\text{m}$) and coarse particulate matter ($\leq 10 \mu\text{m}$), and carbon monoxide (CO). All these pollutants will be measured both indoor and in the ambient environment using air quality monitors. Personal sampling monitors will be used to measure the individual exposure level of particulate matter ($\leq 10 \mu\text{m}$, $\leq 2.5 \mu\text{m}$) and CO at the household level. The eligible participants will be required to wear a personal sampling device for one entire day. The samplers will be kept inside a small fully ventilated, appealing sling bag that they can wear while carrying out the routine household chores (Figure 4). The samplers record the data continuously for 24-hours, and are connected to a portable outlet power source or power bank. The data will be stored in the memory card in the sampler which is then transferred and stored safely in a hard disk later by the research staff. The exposure data will be measured twice during pregnancy, during the second and the third trimester of pregnancy. A tentative schedule for the distribution of the monitors is provided in the supplementary material.²⁶ The ambient air quality monitor will be kept outside the house in a protected environment, within a 5 km radius from the respective participant's house to assess live ambient data on all the three pollutants (PM($\leq 10 \mu\text{m}$, $\leq 2.5 \mu\text{m}$) and CO) for 24 hours. The device will be about five feet above the ground level provided with a shelter to secure the devices from direct sunlight and rain. We will also obtain the readings of the ambient air pollutant levels from the nearest ambient air quality monitoring devices and stations installed by the pollution control board (www.eprolytics.com:9999/#/public_view/kspcb).

Together, we will assess the ambient exposure level of that particular area. This data will be considered as a proxy for ambient data for each household in that area.

Covariates

We will assess socioeconomic and demographic variables, current and previous obstetric history, fuel used for cooking and heating, and the location of the kitchen, using pretested questionnaires. The information on other sources of pollution such as burning of agarbatti/dhoop (Incense sticks); and use of mosquito repellents and candles will be collected. The duration of exposure to indoor as well as outdoor air pollution will be collected.²⁶ Also, trained research staff will record the height and weight of the participants by using SECA 213 (Seca Precision for health) stadiometer and digital Omron HN-283 (Omron Healthcare Co., Ltd.) weighing scale. The scale will be placed on a level ground, the research staff will



Figure 4. Complete set of device holding power bank and ventilated sling bag.

check for a 'zero' reading, and after ensuring that the participant has removed heavy outer clothing and shoes, two readings to the nearest 10 grams will be recorded.

A portable stadiometer will be used for measuring height to the nearest 0.1 cm; measured with the participant standing straight with her feet together, the head plate of the stadiometer will then be pulled down to ensure that it rests on the crown of the head. Blood pressure will be measured using OMRON HEM-7203 (Omron Healthcare Co., Ltd.) automated blood pressure monitor. We will also measure the fasting and 2-hour postprandial blood glucose levels (Oral glucose tolerance test)²⁷ and haemoglobin during the day as a markers of hyperglycaemia and anaemia in pregnant women who complete their 24 weeks of gestation (Table 1). The tests will be conducted by Medall Healthcare Pvt. Ltd. using fully automated analyzer, non-cyanide methods for hemoglobin estimation and glucose will be estimated using GOD-POD, colorimetric method. The procedure has been detailed further in laboratory analysis and sample storage section. The field staff will assess for obstetric morbidity and hospitalization during the monitoring visits.

Household temperature and the status of anaemia will be considered as an intermediate factor between indoor and ambient air pollution and its association with low birth weight.^{28,29} A list of known effect modifiers will be made and collected using a structured questionnaire and later considered during analysis. (Eg; the age of the participant, occupational exposure³⁰).

Season, socioeconomic status, smoking history and exposure to second-hand smoke, traffic roadways, antenatal stress and social support have been considered as confounders.^{31–36} As we cannot identify the relative contribution of tobacco smoke to the indoor or ambient air pollution, we will obtain data using the structured questionnaire. Psychosocial stress data will be collected using the standard Edinburgh Postnatal Depression Scale (EPDS), a widely used self-reporting questionnaire explicitly developed to screen women for perinatal depression.³⁷ EPDS has been validated by Fernandes *et al.* for prenatal depression in South India at a cut-off of ≥ 13 (sensitivity = 100%, specificity = 84.90%, and area under the curve = 0.95).³⁸ Social support will be measured using a questionnaire that has been used and developed by St. John's Research Institute to evaluate a broad range of social support (i.e., emotional, instrumental, informational, and appraisal).³⁹ This questionnaire has a total of 12 items, and each item is scored between 0 (definitely not enough) to 3 (definitely enough). The highest score being 36 means excellent social support and 0 meaning very low social support.

Outcome variables: Birth weight, length and Skinfold thickness of the baby at birth and one year are the outcomes of interest. Feeding practices, morbidity, and child milestone development will be assessed during the follow-up visits (Table 1). The Trivandrum Developmental Scale will be used to measure the developmental milestones.⁴⁰ The 51-items of Trivandrum Development Screening Chart for children of 0–6 y [TDSC (0–6 y)] is a simple, reliable and valid screening tool in the community to identify children between 0–6 years with developmental delay.

For assessment of weight, the baby will be placed naked on the digital SECA 354 weighing scale and readings will be taken to the nearest 0.5 g. For measuring infant length, the baby's head will be held against the end of the head plate of the SECA 417 infantometer, and bringing the foot plate up to the heels ensuring that the feet and knees are flat.

Table 1. Proposed measurements at baseline and follow up in the APPLE study.

Participants	Measurement/tests	Frequency	N	Time points
Pregnant women	PM _{2.5} , PM ₁₀ and CO concentration of both indoor and ambient levels	Two times	516	Once in 2 nd and 3 rd trimester
Pregnant women	1. Blood glucose estimation 2. Haemoglobin 3. Height and weight, Blood Pressure 4. Socio-demographic characteristics, current obstetric history, psychosocial stress, social support	Once	516	During 24–36 weeks of gestation
Newborn	Anthropometry of the child (Skinfold thickness (Biceps, triceps and subscapular) and circumferences (Head, chest, waist, hip and mid-arm circumference)	Once	516	At birth
Infants	Skinfold thickness (Biceps, triceps and subscapular) and circumferences (Head, chest, waist, hip and mid-arm circumference, morbidity and feeding practices, child developmental milestones	Once	516	One year

Chasmors body circumference tape will be used to measure the circumferences. Head circumference will be measured with the baby's head on the side. The tape will be placed on the forehead. Waist circumference will be taken by placing the tape around the abdomen immediately above the umbilicus at the end of expiration. Chest circumference will be measured by placing the tape around the chest at the end of expiration. Mid-upper-arm (MUAC) will be recorded with the arm bent, allowing the measurement to be taken with the baby in its natural position. An ink mark will be made on the anterior and posterior side of the arm to locate the point for biceps and triceps estimation.

Skinfold thickness will be measured on the left side of the body using the Holtain Calipers. Three readings to the nearest 0.2 mm will be taken. For triceps skinfold thickness, the tape will be placed around the upper arm at the level of the mark done at the posterior side while measuring mid-upper arm circumference (MUAC). With the tape in position, a horizontal line will be drawn on the skin at the level of the mark. The point at which the fold is to be measured will then be marked; the skin will be lifted over the posterior surface of the triceps muscle, above the marked point and the callipers will be applied below the fingers. For subscapular skinfold thickness, the inferior angle of the scapula will be identified, and the skin will be marked immediately below the angle. The skinfold will be picked up above the mark, and the calliper jaws are applied below the fingers, at the apex of the fold.⁴¹

Adiposity: We define adiposity as the sum of the skinfold thicknesses such as biceps, triceps and subscapular measuring above the 70th percentiles are classified as at risk to obese and more than 85th percentiles are classified as obese.⁴²

The skinfold thickness equation for body fat composition measurement by Holtain Caliper is correlated and validated against dual-energy x-ray absorptiometry (DEXA) which is a standard gold method for body fat estimation and reported a reasonable validation with DEXA.⁷

Laboratory analysis and sample storage: Medall Healthcare Pvt. Ltd (MEDALL), a centralized and nationally accredited laboratory was engaged to carry out all the tests including haemoglobin and oral glucose tolerance test (OGTT). The pregnant women who completed 24 weeks of gestations will be advised by the research staff to visit the hospital after overnight fasting state for at least 7–8 hours. A trained phlebotomist will draw the fasting and postprandial sample 2 hours following a 75 g oral load of glucose.⁴³ The blood from the fasting sample will be centrifuged within 30 minutes of collection at 3500 rpm for about 5 minutes in a REMI Medico centrifuge C-854/6 portable centrifuge. The sample will be transferred into a cold box at 2–8°C and will be transferred to the central laboratory, where assays will be carried out. An aliquot of plasma will be made by centrifuging the fasting Sodium Fluoride tube and haemoglobin processed EDTA samples for 3500 rpm for 5 min. The remaining sediment of packed red cells will be centrifuged at high speed at 4500 rpm for 5–10 min to extract the buffy coat samples. The plain tube containing 6 ml of blood will be made to stand for 40 minutes to clot then centrifuged at a speed of 4500 rpm for 5–10 minutes. In total six aliquots of serum will be separated from 6 ml of a plain blood sample each 0.5 ml. 4 aliquots of plasma of 0.5 ml each, 3 aliquots of buffy coats (0.1–0.5 ml) will be stored for the future analysis. The aliquots will be stored in a biorepository at the research centre (Indian Institute of Public Health, Public Health Foundation of India-Bangalore), wherein the samples will be stored in a stepwise manner (2–8°C, –20°C deep freezer and –80°C deep freezer) for future analysis. The samples are planned to be stored long term will be moved to a –80°C, ultra low deep freezer.

Sample size: A research study by Kalpana Balakrishna *et al.* showed that a population-weighted mean of annual PM_{2.5} exposure in India has increased from 59.8 µg/m³ to 79.9 µg/m³ (1990–2016). The study also reported that more than 80% of the Indian population were exposed to PM_{2.5} of more than 40 µg/m³. Delhi has the highest populated weighted mean of PM_{2.5} (>150 µg/m³) followed by Uttar Pradesh, Bihar and Haryana. We plan to recruit 516 pregnant women over a period of one year with a hypothesized (based on the evidence in India) 20% of LBW in this population,⁴⁴ accounting for the design effect (DEFF) of (for cluster surveys- 1.5, for 95% confidence interval), the sample size required is 369. Further, due to the transient nature of pregnant women from the slum population, the sample size required is 516 accounting for 40% loss to follow-up.⁴⁵

The sample size has been estimated using a formula, $n = Z^2 P(1 - P)/d^2$; where n is the sample size, Z is the statistic corresponding to the confidence interval, P is the expected prevalence and d is the precision. The sample size of 246 has been estimated using 20% of expected prevalence of LBW at 95% confidence interval and 5% precision. The two zones of the study area considered as a cluster and accounting for the design effect (DE) of 1.5 the sample size came out to be 369. Further, due to the transient nature of pregnant women from the slum population, the sample size required is 516 accounting for 40% loss to follow-up. We conducted a post-hoc power analysis using G*Power software. Based on our final sample size of 516 participants, the analysis confirms a statistical power of 98% (at an alpha level of 0.05) to detect effect size of $P = 0.20$. This indicates that the study is sufficiently powered to identify significant associations even if the differences in exposure levels are relatively small.

Data sources and measurements

The study instruments used in this study have been developed by research team using previous research finding. The questionnaire has been piloted and validated before administration in the field. The research staff will administer the questionnaire, conduct blood investigation, and record the anthropometric measurements at the health centre. The air monitoring device with power bank will be distributed to pregnant women by the field staff of the respective area and taken back once the 24 hours estimation is done. The post-monitoring assessment will be done during this period for the pregnant woman. The post-monitoring assessment will be performed using a structured questionnaire assessing the cooking practices and activity of the participants during the monitoring period. The enrolled pregnant women due for delivery will be tracked through periodic phone calls and scheduling follow-up visits.

Ambient and indoor Air quality monitor: We will use the VAYUSESNE (later renamed as VAYUCARE) device for assessing the outdoor and indoor air pollutants. This is designed by Ambience Monitoring India private limited based in Delhi. The selling partner of the company products is I love Clean Air (<http://www.ilcalab.com>). It is a portable device that can operate for 24 hours with 10,000mAh external power bank (Figure 4). The devices provide real-time monitoring of particulate pollution ($\leq 10 \mu\text{m}$, $\leq 2.5 \mu\text{m}$) and CO. It has air sensors to produce details on air quality. The data will be saved in an SD card. The downloaded data can be viewed in comma-separated values (CSV) and Excel Microsoft Office Open XML Format Spread-sheet file (XLSX) format. The monitors used in this study were locally calibrated against a collocated E-BAM (Beta Attenuation Monitor), a Federal Equivalent Method (FEM) standard instrument. We generated device-specific scatter plots which showed $R^2 > 0.88$. Following best practices, we derived and applied device-specific linear regression correction equations to all particulate matter (PM) data to align the low-cost sensor readings with the E-BAM standard (Figures 1 and 2, Supplementary File). In addition to the E-BAM calibration, we applied a correction factor to account for hygroscopic growth of particles at high humidity levels, ensuring that the reported concentrations represent dry particle mass.

Bias

Specific attention will be paid to limit bias by controlling for confounding, minimizing selection bias and measurement errors. In order to prevent any differential misclassification of outcome, the data from laboratory investigations and air pollutant estimation is accessed only by the key research staff who are not involved in data collection. The field team will be trained and certified on anthropometric measurements by St. John's Research Institute and any possible measurement error will be minimized. Accuracy and inter-observer reliability of their measurements will be assessed at the outset and every subsequent six months. To control for confounding, information about the potential confounders (Table 1) will be obtained and controlled during the analysis stage. As this is a pregnancy cohort study, outcome measures obtained only from the live born children are considered (live birth bias) and reason for pregnancy loss and neonatal death are collected from those whose child does not survive from primary caregiver and in case of absence of primary caregiver responses will be collected from the secondary caregiver.

Statistical methods: Descriptive statistical estimates of the individual exposure will be reported. Frequencies and percentages will be reported for all the variables. The exposure data distribution will be checked for normality. The mean and standard deviation will be reported for quantitative variables. The minimum and maximum level of exposure will be reported. The prevalence of low birth weight will be calculated. The fasting and postprandial glucose will be categorized based on the WHO recommended cut off levels (fasting glucose of ≥ 92 and 2-hr values of ≥ 153) are considered as hyperglycemia.⁴⁷ The anaemia status will be categorized as normal when the Hemoglobin (Hb) is $>11 \text{ gm/dl}$, mild anaemia when Haemoglobin is $10\text{--}10.9 \text{ gm/dl}$, moderate anaemia when Hemoglobin is $7\text{--}9.9 \text{ gm/dl}$ and severe when the Hemoglobin level is less than 7 gm/dl .^{48,49} The outcomes (birth weight and adiposity) will be modelled both as a continuous and binary indicator variable. The exposure of interest ($\text{PM}_{2.5}$, PM_{10} and CO) will be taken as a continuous scale measure for analysis. The continuous measured exposure value will be categorized in terms of quartiles and the lower quartile of the exposure will be taken as the reference category for each pollutant for further analysis. The risk ratio with confidence interval will be reported. We will also report the dose-response relationship between maternal exposure to indoor and ambient air pollutants ($\text{PM}(\leq 10 \mu\text{m}, \leq 2.5 \mu\text{m})$ and CO) and birth weight will be assessed through sensitivity analysis.^{50,51} Mediation analysis will be used to test the hypothesized causal chain for third and fourth objectives; involving low birth weight as an intermediate in the association between air pollution and adiposity. Further we also explore sensitivity analyses stratified by personal factors (e.g., age, socio economic status and occupational history of exposure).

Dissemination of information: The findings of the study will assist the efforts of the Government to counter climate change. The data from the study will pave way for future research and policy-making agendas of the government. We have engaged key stakeholders like health officers and Slum board official to increase their sense of awareness towards the problem of air pollution towards health. Through membership of our National Advisory Group and periodic input,

will ensure the wider dissemination of our findings. The findings will be disseminated to fellow researchers through conference presentations, publications in the peer reviewed journals and through reports to health authorities.

Study status: The baseline assessments have been started at two health facilities of the study area. The study tool has been piloted in the field and necessary changes have been made. We have completed the process of calibration of air quality devices before we introduce in the field level. The calibration process has been done against the standard E-BAM monitor. We have completed at birth and one year follow up in 229 & 82 infants respectively.

Discussion

There is a lack of standardized data for the confluence of risk factors including the levels and roles of specific pollutants and how they are associated with low birth weight and adiposity in India. By collecting high-quality prospective data on exposures in pregnant women, this study can provide insights into the environmental causes of low birth weight and obesity in childhood. The results from our study may provide evidence regarding the adverse effects of air pollution in pregnancy, and thereby can help in improving the neonatal and child health outcomes. The results can inform policy regarding limiting the air pollution and in the design of interventions for use in future studies. While this study focuses on environmental challenges of urban slums, further comparative research involving non-urban settings would fully quantify the scale of environmental inequity.

Data availability

Underlying data

No underlying data are associated with this article.

Extended data

Figshare: Supplementary file has table and three figures. Table has detailed information on exposure, risk factors, confounder, effect modifier, intermediate and outcome. [Figures 1 and 2](#) has directed acyclic graph for objective 1 and 2. The [Figure 3](#) shows ongoing monitoring stations in Bengaluru which was installed by State Pollution Control Board. <https://doi.org/10.6084/m9.figshare.11980482.v1>²⁶

Figshare: Supplementary file including Appendices 1 and 2. <https://doi.org/10.6084/m9.figshare.11980482.v1>⁵²

Acknowledgements

We sincerely thank Department of Health and Family Welfare, Government of Karnataka (DHFV, GoK) and The Bruhat Bengaluru Mahanagara Palike (BBMP) for granting us permission to conduct the study and constant support. We thank hospitals under DHFW, GoK, Superintendents, Medical Officers, doctors and all the support staffs of the department for the support in ongoing study. We sincerely thank Karnataka State Pollution Control Board for permitting us for the external calibration process of air quality devices and Dr. Megha Takur for her support in the internal calibration process under standard circumstances. We sincerely thank Dr Jasper V. Been, Consultant Neonatologist, University Medical Centre Rotterdam, Rotterdam, Zuid Holland, Netherlands for thorough review and helpful inputs for the protocol. We thank Dr Megha Thakur for proofreading the manuscript. Our sincere thanks to Dr Suresh Shapeti and Mr Ramesh for facilitating the administrative support and coordination. We thank Gurulingaiah, Kiran, Keerthi Deshpande, Maithili Karthik, Sindhu Gowda and Meena KV for their support in carrying out research activities at the field level.

References

- Lamichhane DK, Leem JH, Lee JY, *et al.*: **A meta-analysis of exposure to particulate matter and adverse birth outcomes.** *Environ Health Toxicol.* 2015; **30**: e2015011. [PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
- Checkley W, Pollard SL, Siddharthan T, *et al.*: **Managing threats to respiratory health in urban slums.** *Lancet Respir Med.* 2016; **4**(11): 852–854. [PubMed Abstract](#) | [Publisher Full Text](#)
- Landrigan PJ, Fuller R, Acosta NJR, *et al.*: **The Lancet Commission on pollution and health.** *Lancet.* 2018; **391**(10119): 462–512. [Publisher Full Text](#)
- Ezzati M, Kammen DM: **The health impacts of exposure to indoor air pollution from solid fuels in developing countries: knowledge, gaps, and data needs.** *Environ Health Perspect.* 2002; **110**(11): 1057–1068. [PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
- Arku RE, Birch A, Shupler M, *et al.*: **Characterizing exposure to household air pollution within the Prospective Urban Rural Epidemiology (PURE) study.** *Environ Int.* 2018; **114**: 307–317. [PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
- INDIA PROFILE: *Institute for Health Metrics and Evaluation.* 2016; **22**: 323–341. [PubMed Abstract](#)
- Stieb DM, Chen L, Eshoul M, *et al.*: **Ambient air pollution, birth weight and preterm birth: a systematic review and meta-analysis.** *Environ Res.* 2012; **117**: 100–111. [PubMed Abstract](#) | [Publisher Full Text](#)

8. Dadvand P, Parker J, Bell ML, *et al.*: **Maternal exposure to particulate air pollution and term birth weight: a multi-country evaluation of effect and heterogeneity.** University of British Columbia; 2015.
9. Shah PS, Balkhair T: **Knowledge Synthesis Group on Determinants of Preterm/LBW births: Air pollution and birth outcomes: a systematic review.** *Environ Int.* 2011; **37**(2): 498–516. [PubMed Abstract](#) [Publisher Full Text](#)
10. Tielsch JM, Katz J, Thulasiraj RD, *et al.*: **Exposure to indoor biomass fuel and tobacco smoke and risk of adverse reproductive outcomes, mortality, respiratory morbidity and growth among newborn infants in south India.** *Int J Epidemiol.* 2009; **38**(5): 1351–1363. [PubMed Abstract](#) [Publisher Full Text](#)
11. Balakrishnan K, Ghosh S, Thangavel G, *et al.*: **Exposures to fine particulate matter (PM_{2.5}) and birthweight in a rural-urban, mother-child cohort in Tamil Nadu, India.** *Environ Res.* 2018; **161**: 524–531. [PubMed Abstract](#) [Publisher Full Text](#)
12. Sun X, Luo X, Zhao C, *et al.*: **The associations between birth weight and exposure to fine particulate matter (PM_{2.5}) and its chemical constituents during pregnancy: A meta-analysis.** *Environ Pollut.* 2016; **211**: 38–47. [PubMed Abstract](#) [Publisher Full Text](#)
13. Atmospheric Pollution and Human Health in an Indian Megacity. [Reference Source](#)
14. Janssen BG, Munters E, Pieters N, *et al.*: **Placental mitochondrial DNA content and particulate air pollution during *in utero* life.** *Environ Health Perspect.* 2012; **120**(9): 1346–1352. [PubMed Abstract](#) [Publisher Full Text](#) [Free Full Text](#)
15. van den Hooven EH, Pierik FH, de Kluizenaar Y, *et al.*: **Air pollution exposure and markers of placental growth and function: the generation R study.** *Environ Health Perspect.* 2012; **120**(12): 1753–1759. [PubMed Abstract](#) [Publisher Full Text](#) [Free Full Text](#)
16. Damasceno-Rodrigues NR, Veras MM, Negri EM, *et al.*: **Effect of pre- and postnatal exposure to urban air pollution on myocardial lipid peroxidation levels in adult mice.** *Inhal Toxicol.* 2009; **21**(13): 1129–1137. [PubMed Abstract](#) [Publisher Full Text](#)
17. Salam MT, Millstein J, Li YF, *et al.*: **Birth outcomes and prenatal exposure to ozone, carbon monoxide, and particulate matter: results from the Children's Health Study.** *Environ Health Perspect.* 2005; **113**(11): 1638–1644. [PubMed Abstract](#) [Publisher Full Text](#) [Free Full Text](#)
18. Hales CN, Barker DJ: **The thrifty phenotype hypothesis.** *Br Med Bull.* 2001; **60**(1): 5–20. [Publisher Full Text](#)
19. Sharma D, Shastri S, Sharma P: **Intrauterine Growth Restriction: Antenatal and Postnatal Aspects.** *Clin Med Insights Pediatr.* 2016; **10**: 67–83. [PubMed Abstract](#) [Publisher Full Text](#)
20. Ornoy A: **Prenatal origin of obesity and their complications: Gestational diabetes, maternal overweight and the paradoxical effects of fetal growth restriction and macrosomia.** *Reprod Toxicol.* 2011; **32**(2): 205–212. [PubMed Abstract](#) [Publisher Full Text](#)
21. Medici M, Timmermans S, Visser W, *et al.*: **Maternal thyroid hormone parameters during early pregnancy and birth weight: the Generation R Study.** *J Clin Endocrinol Metab.* 2013; **98**(1): 59–66. [PubMed Abstract](#) [Publisher Full Text](#)
22. Rosa MJ, Pajak A, Just AC, *et al.*: **Prenatal exposure to PM_{2.5} and birth weight: A pooled analysis from three North American longitudinal pregnancy cohort studies.** *Environ Int.* 2017; **107**: 173–180. [PubMed Abstract](#) [Publisher Full Text](#) [Free Full Text](#)
23. Guardino CM, Schetter CD, Saxbe DE, *et al.*: **Diurnal salivary cortisol patterns prior to pregnancy predict infant birth weight.** *Health Psychol.* 2016; **35**(6): 625–633. [PubMed Abstract](#) [Publisher Full Text](#) [Free Full Text](#)
24. Visentin S, Grumolato F, Nardelli GB, *et al.*: **Early origins of adult disease: low birth weight and vascular remodeling.** *Atherosclerosis.* 2014; **237**(2): 391–399. [PubMed Abstract](#) [Publisher Full Text](#)
25. Sinding M, Peters DA, Frøkjær JB, *et al.*: **Prediction of low birth weight: Comparison of placental T2* estimated by MRI and uterine artery pulsatility index.** *Placenta.* 2017; **49**: 48–54. [PubMed Abstract](#) [Publisher Full Text](#)
26. Shriyan P, Babu GR, Ravi D, *et al.*: **Ambient and Indoor Air Pollution in Pregnancy and the risk of Low birth weight and Ensuing Effects in Infants (APPLE): A cohort study in Bangalore, South India.** *Figshare.* Figure. 2020. [Publisher Full Text](#), [3](#)
27. Naylor CD: **Diagnosing gestational diabetes mellitus. Is the gold standard valid?** *Diabetes Care.* 1989; **12**(8): 565–572. [PubMed Abstract](#) [Publisher Full Text](#)
28. Bobak M: **Outdoor air pollution, low birth weight, and prematurity.** *Environ Health Perspect.* 2000; **108**(2): 173–176. [PubMed Abstract](#) [Publisher Full Text](#) [Free Full Text](#)
29. Stanković A, Nikić D, Nikolić M: **Relationship between exposure to air pollution and occurrence of anemia in pregnancy.** *Facta Univ Ser Med Biol.* 2006; **13**(1): 54–57. [Reference Source](#)
30. Savitz DA, Whelan EA, Kleckner RC: **Effect of parents' occupational exposures on risk of stillbirth, preterm delivery, and small-for-gestational-age infants.** *Am J Epidemiol.* 1989; **129**(6): 1201–1218. [PubMed Abstract](#) [Publisher Full Text](#)
31. Wang X, Ding H, Ryan L, *et al.*: **Association between air pollution and low birth weight: a community-based study.** *Environ Health Perspect.* 1997; **105**(5): 514–520. [PubMed Abstract](#) [Publisher Full Text](#) [Free Full Text](#)
32. Shu XO, Hatch MC, Mills J, *et al.*: **Maternal smoking, alcohol drinking, caffeine consumption, and fetal growth: results from a prospective study.** *Epidemiology.* 1995; **6**(2): 115–120. [PubMed Abstract](#) [Publisher Full Text](#)
33. Ritz B, Wilhelm M, Hoggatt KJ, *et al.*: **Ambient air pollution and preterm birth in the environment and pregnancy outcomes study at the University of California, Los Angeles.** *Am J Epidemiol.* 2007; **166**(9): 1045–1052. [PubMed Abstract](#) [Publisher Full Text](#)
34. Jalaludin B, Mannes T, Morgan G, *et al.*: **Impact of ambient air pollution on gestational age is modified by season in Sydney, Australia.** *Environ Health.* 2007; **6**(1): 16. [PubMed Abstract](#) [Publisher Full Text](#) [Free Full Text](#)
35. Lee BE, Ha EH, Park HS, *et al.*: **Exposure to air pollution during different gestational phases contributes to risks of low birth weight.** *Hum Reprod.* 2003; **18**(3): 638–643. [PubMed Abstract](#) [Publisher Full Text](#)
36. Ha S, Hu H, Roussos-Ross D, *et al.*: **The effects of air pollution on adverse birth outcomes.** *Environ Res.* 2014; **134**: 198–204. [PubMed Abstract](#) [Publisher Full Text](#) [Free Full Text](#)
37. Babu GR, Murthy G, Deepa R, *et al.*: **Maternal antecedents of adiposity and studying the transgenerational role of hyperglycemia and insulin (MAASTHI): a prospective cohort study: Protocol of birth cohort at Bangalore, India.** *BMC Pregnancy Childbirth.* 2016; **16**(1): 311. [PubMed Abstract](#) [Publisher Full Text](#) [Free Full Text](#)
38. Fernandes MC, Srinivasan K, Stein AL, *et al.*: **Assessing prenatal depression in the rural developing world: a comparison of two screening measures.** *Arch Womens Ment Health.* 2011; **14**(3): 209–216. [PubMed Abstract](#) [Publisher Full Text](#)
39. Babu GR, Murthy GVS, Reddy Y, *et al.*: **Small for gestational age babies and depressive symptoms of mothers during pregnancy: Results from a birth cohort in India [version 1; referees: 1 approved, 1 approved with reservations].** *Wellcome Open Res.* 2018; **3**: 76. [Publisher Full Text](#)
40. Nair MK, Nair GS, George B, *et al.*: **Development and validation of Trivandrum Development Screening Chart for children aged 0–6 years (TDSC (0–6)).** *Indian J Pediatr.* 2013; **80**(Suppl 2): 248–255. [Publisher Full Text](#)
41. Veena S: **Cognitive performance during childhood and early adolescence in India: relationships to birth size, maternal nutrition during pregnancy and postnatal growth.** University of Southampton. 2015. [Reference Source](#)
42. Khadilkar A, Mandlik R, Chiplonkar S, *et al.*: **Reference centile curves for triceps skinfold thickness for Indian children aged 5 to 17 years and cut offs for predicting risk of childhood hypertension: a multi-centric study.** *Indian Pediatr.* 2015; **52**(8): 675–680. [PubMed Abstract](#) [Publisher Full Text](#)
43. Agarwal MM: **Gestational diabetes mellitus: Screening with fasting plasma glucose.** *World J Diabetes.* 2016; **7**(14): 279–289. [PubMed Abstract](#) [Publisher Full Text](#) [Free Full Text](#)
44. Chellani R, Lopamudra P, Kulkarni P: **Incidence of Low-Birth-Weight in India: Regional Variations and Socio-Economic Disparities.** *J Health Dev.* 2007; **3**(1): 148–162. [Reference Source](#)
45. Dean A, Sullivan K, Soe M: **OpenEpi: Open Source Epidemiologic Statistics for Public Health, Version 2.3. 1.** 2011. [cited 2310 10]. 2013. [Reference Source](#)

46. Vervoort JWP: **Healthy low energy redesigns for schools in Delhi.** *Eindhoven University of Technology*. 2018.
[Reference Source](#)
47. World Health Organization: **Diagnostic Criteria and Classification of Hyperglycaemia First Detected in Pregnancy.** 2013.
48. World Health Organization: **Nutritional anaemias: tools for effective prevention and control.** 2017.
[Reference Source](#)
49. World Health Organization: **Haemoglobin concentrations for the diagnosis of anaemia and assessment of severity.** 2011.
[Reference Source](#)
50. Greenland S, Daniel R, Pearce N: **Outcome modelling strategies in epidemiology: traditional methods and basic alternatives.** *Int J Epidemiol.* 2016; **45**(2): 565–575.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
51. Stata Corporation: **Stata Statistical Software Release 7.0: Reference Q-St.** Stata Corporation; 2001.
52. Shriyan P, Babu GR, Ravi D, *et al.*: **Ambient and Indoor Air Pollution in Pregnancy and the risk of Low birth weight and Ensuing Effects in Infants (APPLE): A cohort study in Bangalore, South India.** *figshare*. Figure. 2020; **3**.
[Publisher Full Text](#)

Open Peer Review

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Version 3

Reviewer Report 10 January 2025

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? **Francisco Jablinski Castelhana** 

¹ Federal University of Rio Grande do Norte, Rio Grande do Norte, Brazil

² Federal University of Rio Grande do Norte, Rio Grande do Norte, Brazil

Dear authors, I appreciate the opportunity to review your research.

I was impressed with the amount of data you collected, from both environmental and physiological variables, however, I must say that the socioeconomical variables are the ones that come to my attention.

According to several researches I read in the past years, there's a clear relationship between air pollution and socioeconomical factors in India, due to the use of coal and other biomass fuels on cooking as you mentioned in introduction, so this is a clear case of environmental justice. Despite the fact that you studied only the slums of Bangalore perhaps a comparison with other parts of the city that are not slums could be interesting.

I also believe that the adiposity is not quite well explained. Recommend that you take more time to explain how air pollution could mediate the adiposity situation on the babies.

I strongly recommend to remake the map of figure 2 to make it more scientific by using any GIS software, by doing this, you should add cartographical elements such as scale, wind rose and others.

Besides that, I recommend the reading of the following paper that could also help you with the LBW and air pollution matter

Is the rationale for, and objectives of, the study clearly described?

Partly

Is the study design appropriate for the research question?

Yes

Are sufficient details of the methods provided to allow replication by others?

Yes

Are the datasets clearly presented in a useable and accessible format?

Yes

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Geography, Air pollution, Environmental sciences

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard, however I have significant reservations, as outlined above.

Author Response 09 Sep 2026

Giridhara R Babu

We thank the reviewer for the comments. We have responded to each of your queries and hope our responses address your concerns.

1. According to several researches I read in the past years, there's a clear relationship between air pollution and socioeconomical factors in India, due to the use of coal and other biomass fuels on cooking as you mentioned in introduction, so this is a clear case of environmental justice. Despite the fact that you studied only the slums of Bangalore perhaps a comparison with other parts of the city that are not slums could be interesting.

Response: We thank the reviewer for this observation regarding environmental justice and socioeconomic determinants of air pollution. We agree that a comparison between urban slums and urban or rural residential areas would provide a strong illustration of the disparities. However, the study goal was to characterise the specific exposure risks among the most vulnerable, high-density urban settlements. While the city-wide comparison is beyond the scope of the current paper, we have added a statement in the discussion section acknowledging the need for future research comparing the exposure risks with non-slum areas, which would further highlight environmental justice, as the reviewer correctly identifies (Page No. 13, Line No. 574- 577).

2. I also believe that the adiposity is not quite well explained. Recommend that you take more time to explain how air pollution could mediate the adiposity situation on the babies.

Response: We thank the reviewer for the opportunity to further elaborate on the biological mechanisms linking air pollution to infant adiposity. We have expanded the introduction section to provide more detail on how air pollution results in infant adiposity (Page No. 3, Line No. 104-109). The expanded section includes the following changes: Maternal exposure to air pollutants induces placental dysfunction and systemic inflammation, which disrupts the fetal energy homeostasis and adipokine regulation [1, 2]. These triggers impair insulin signaling and promote lipidomic changes that 'program' the fetus for altered adipose tissue development, significantly increasing long-term obesity risks [3, 4].

3. I strongly recommend to remake the map of figure 2 to make it more scientific by

using any GIS software, by doing this, you should add cartographical elements such as scale, wind rose and others.

Response: We thank the reviewer for the constructive feedback on Figure 2. We agree that professional cartographic elements enhance the scientific value of the map. Accordingly, we have recreated the figure using QGIS. We hope this resolves your concern.

4. Besides that, I recommend the reading of the following paper that could also help you with the LBW and air pollution matter

Response: We thank the reviewer for this recommendation. We are very interested in incorporating this relevant literature into our discussion; however, it appears that the specific titles or links were not included in the review comments we received. We have added additional literature on the link between air pollution and LBW in Introduction to ensure our manuscript is up to date (Page No-3, Line No:114).

Competing Interests: Authors declare that there is no competing interests.

Reviewer Report 24 April 2024

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Jiawen Liao

¹ University of Southern California, Los Angeles, California, USA

² University of Southern California, Los Angeles, California, USA

Review for manuscript

I want to thank authors for this opportunity for me to review this protocol. I reviewed this manuscript and comments from reviewer Dr. Lu, Dr. Malmqvist, Dr. Balakrishnan and Dr. Allen and Dr. Marshall.

My comments are listed in below:

1. Figure 2 showing the map looks like they are scaled. I would recommend authors to provide better quality maps, preferably using ArcGIS or QGIS software.

2. The statements: "The skinfold thickness equation for body fat composition measurement by Holtain Caliper is correlated and validated against dual-energy x-ray absorptiometry (DEXA) which is a standard gold method for body fat estimation and reported a reasonable validation with DEXA." Need a reference.

3. Blood sample collection: from the protocol, it looks like authors are just collecting fasting blood samples from pregnant women in a hospital. Authors shouldn't mention that "assays will be carried out" in the manuscript, because the activity is only involved sample collection.

4. Blood sample collection time: authors should provide some more details at what point of pregnancy the blood is collected, at first, second or third trimester?

Is the rationale for, and objectives of, the study clearly described?

Yes

Is the study design appropriate for the research question?

Partly

Are sufficient details of the methods provided to allow replication by others?

Partly

Are the datasets clearly presented in a useable and accessible format?

Yes

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: household air pollution, exposure assessment, environmental epidemiology

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard, however I have significant reservations, as outlined above.

Author Response 09 Sep 2026

Giridhara R Babu

We thank the reviewer for their comments. We have responded to each of the comments and hope our responses address your concern.

1. Figure 2 showing the map looks like they are scaled. I would recommend authors to provide better quality maps, preferably using ArcGIS or QGIS software.

Response: We thank the reviewer for this recommendation. As suggested, we have replaced Figure 2 with a high-resolution map generated using QGIS, ensuring better scaling and visual quality.

2.The statements: "The skinfold thickness equation for body fat composition measurement by Holtain Caliper is correlated and validated against dual-energy x-ray absorptiometry (DEXA) which is a standard gold method for body fat estimation and reported a reasonable validation with DEXA." Need a reference.

Response: We thank the reviewer for noting this. We have now included the appropriate

references to support the validation of Holtain Caliper skinfold equations against DEXA.

3. Blood sample collection: from the protocol, it looks like authors are just collecting fasting blood samples from pregnant women in a hospital. Authors shouldn't mention that "assays will be carried out" in the manuscript, because the activity is only involved sample collection.

Response: We thank the reviewer for this clarification. The research institution had an MoU with the NABL-accredited laboratory to carry out the laboratory investigations planned under the project. To clarify the scope of the work, laboratory investigations—including hemoglobin levels and the 2-hour OGTT (fasting and postprandial)—were performed by Medall Healthcare Pvt. Ltd (MEDALL). These investigations were an integral part of the study.

4. Blood sample collection time: authors should provide some more details at what point of pregnancy the blood is collected, at first, second or third trimester?

Response: We thank the reviewer for pointing this out. We have clarified the timing of the blood collection in the revised manuscript. Blood samples were collected from pregnant women upon the completion of 24 weeks of gestation. These details have been added to the 'Laboratory analysis and sample storage' section (Page 11, Line 441) under the Methods section.

Competing Interests: Authors declare that there is no competing interests

Reviewer Report 01 April 2024

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Chan Lu

¹ Central South University, Changsha, Hunan, China

² Central South University, Changsha, Hunan, China

This study is of significance. However, there are some major problems especially for the discussion section.

I have some comments as follows:

1. In the abstract, the definition of "particulate matter (<10 µm, <2.5 µm)" is wrong. Should be "PM10 (<=10 µm), PM2.5 (<=2.5 µm)".
2. The structure is strange. In the introduction, "objectives: We aim to ..." This sentence should

move to the abstract.

3. Figure 2 is strange. Please re-draw this map and make it more scientific.

4. The results is too simple. Please do more analysis especially including some sensitivity analysis stratified by personal factors.

5. The discussion section is too short. This is a research article, which should include a relatively long and rich discussion comparing your results with other studies.

6. I suggest the authors adding some important references in the introduction and/or discussion as follows:

1. Lu C, et al., 2020 [Ref 1]

2. Lu C, et al., 2019 [Ref 2]

3. Lu C, et al., 2021 [Ref 3]

References

1. Lu C, Zhang W, Zheng X, Sun J, et al.: Combined effects of ambient air pollution and home environmental factors on low birth weight. *Chemosphere*. 2020; **240**: 124836 [PubMed Abstract](#) | [Publisher Full Text](#)

2. Lu C, Cao L, Norbäck D, Li Y, et al.: Combined effects of traffic air pollution and home environmental factors on preterm birth in China. *Ecotoxicol Environ Saf*. 2019; **184**: 109639 [PubMed Abstract](#) | [Publisher Full Text](#)

3. Lu C, Deng M, Norbäck D, Liu Z, et al.: Effect of outdoor air pollution and indoor environmental factors on small for gestational age. *Building and Environment*. 2021; **206**. [Publisher Full Text](#)

Is the rationale for, and objectives of, the study clearly described?

Yes

Is the study design appropriate for the research question?

Yes

Are sufficient details of the methods provided to allow replication by others?

Yes

Are the datasets clearly presented in a useable and accessible format?

Yes

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: environment and health

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard, however I have significant reservations, as outlined above.

Author Response 09 Sep 2026

Giridhara R Babu

We thank the reviewer for the comments. We have responded to each query and hope our

responses address your concerns.

1. In the abstract, the definition of "particulate matter (<10 µm, <2.5 µm)" is wrong. Should be "PM10 (≤10 µm), PM2.5 (≤2.5 µm)".

Response: We thank the reviewer for this careful reading and for identifying the notational error. We have corrected the definitions to PM10(≤10 µm) and PM2.5(≤2.5 µm). These changes have been implemented throughout the abstract and the entire manuscript to ensure technical accuracy and consistency.

2. The structure is strange. In the introduction, "objectives: We aim to ..." This sentence should move to the abstract.

Response: We thank the reviewer for the suggestion regarding the placement of the study objectives. To clarify, the objectives are currently stated in the Abstract (as is standard) at the conclusion of the Introduction to provide a clear transition into the methods. We have reviewed the structure of the Introduction to ensure it follows an academic flow, ending with a concise statement of our aims. If the reviewer feels the current phrasing is redundant, we would be happy to refine the wording to better suit the flow of the manuscript.

3. Figure 2 is strange. Please re-draw this map and make it more scientific.

Response: We thank the reviewer for the feedback regarding Figure 2. We have completely redesigned the study site map to adhere to professional cartographic standards. The updated figure now includes a clear hierarchical layout (National -State -District).

4. The results is too simple. Please do more analysis especially including some sensitivity analysis stratified by personal factors.

Response: We thank the reviewer for the suggestion to deepen the analytical approach. We would like to clarify that this manuscript is submitted as a Study Protocol, intended to detail the design, methodology, and planned procedures of the APPLE cohort. As such, it does not include the full result sets from the implementation phase. However, we agree that stratified analysis is crucial. We have updated the "Planned Statistical Analysis" section to explicitly include sensitivity analyses stratified by personal factors (e.g., age, gender, pre-existing health conditions, and socio-economic status) to ensure that the research outputs are robust and account for individual-level variability (Page No-13, Line No:593-595).

5. The discussion section is too short. This is a research article, which should include a relatively long and rich discussion comparing your results with other studies.

Response: We thank the reviewer for this comment and for the opportunity to clarify the scope of this manuscript. We would like to highlight that this submission is a Study Protocol rather than an original research article. In accordance with the reporting guidelines for protocols (e.g., SPIRIT), the discussion section is intended to focus on the rationale, feasibility, and potential impact of the study design rather than the interpretation of results.

6. I suggest the authors adding some important references in the introduction and/or discussion as follows: 1. Lu C, et al., 2020 [Ref 1] 2. Lu C, et al., 2019 [Ref 2] 3. Lu C, et al., 2021 [Ref 3]

Response: We thank the reviewer for suggesting these relevant references. We have reviewed the work, and agree that they provide valuable context regarding the health impacts of particulate matter. We have now incorporated these citations into the Introduction to further strengthen the rationale for the APPLE cohort study and to highlight how our protocol builds upon existing research in this field (Page No-3, Line no-105-110).

Competing Interests: Authors declare that there is no competing interests

Reviewer Report 28 March 2024

<https://doi.org/10.21956/wellcomeopenres.17269.r77103>

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Ebba Malmqvist

¹ Lund University, Lund, Sweden

² Lund University, Lund, Sweden

Thank you for providing info on an important cohort and to contribute to an important knowledge-gap in deprived neighborhoods. I do, however, have some concern on methodology. My first caveat is to assess exposure by 2 times 24 hour measurements. I think this needs to be complemented by questionnaires both in general and during the exposure assessment. This should include questions on ventilation (number of rooms, doors and windows) and if cooking is done outside, in the same room as sleeping or in a separate room. Type of fuel used and time for cooking and who performs the task.

Secondly, I am a bit worried to see traffic as a confounder because it is part of the exposure and adjusting for it will remove effects possibly.

Thirdly, the monitors used has a statement that they have been used for research but that link is broken so impossible to assess how good they are. Low-cost sensors normally has to be validated against reference instruments. They also need to be adjusted for humidity in a non-linear way.

I am also a bit worried, as the other reviewer, about statistical power. If there are no big difference in exposure it might be even harder to find evidence.

Is the rationale for, and objectives of, the study clearly described?

Yes

Is the study design appropriate for the research question?

Partly

Are sufficient details of the methods provided to allow replication by others?

Partly

Are the datasets clearly presented in a useable and accessible format?

Partly

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: I am an epidemiologist and environmental health scientists with a specific focus on air pollution and climate change.

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard, however I have significant reservations, as outlined above.

Author Response 09 Sep 2026

Giridhara R Babu

We thank the reviewer for their comments. We have responded to each of your comments and made changes in the manuscript. We hope the revision addresses your concerns.

My first caveat is to assess exposure by 2 times 24 hour measurements. I think this needs to be complemented by questionnaires both in general and during the exposure assessment. This should include questions on ventilation (number of rooms, doors and windows) and if cooking is done outside, in the same room as sleeping or in a separate room. Type of fuel used and time for cooking and who performs the task.

Response: We thank the reviewer for this comment and agree that behavioral and environmental context is vital for interpreting exposure data. As described in the "Data sources and measurements" section of the manuscript, we have already collected and integrated this information. Specifically, our assessment included ventilation and housing (Number of rooms/windows, kitchen-specific ventilation, and whether windows are kept open during cooking), cooking dynamics (location of cooking, frequency of cooking, and the specific menu/food types prepared during the monitoring period), time and activity during the monitoring period, identification of the primary cook and the duration of cooking tasks. We also recorded information on secondary exposure sources, such as tobacco smoke from other family members. These qualitative data points were collected via both general and post-monitoring questionnaires to complement the 24-hour exposure measurements

Secondly, I am a bit worried to see traffic as a confounder because it is part of the exposure and adjusting for it will remove effects possibly.

Response: We thank the reviewer for this important observation regarding the inclusion of traffic as a confounder. We agree that since ambient traffic contributes significantly to the total personal exposure, adjusting for it could lead to over-adjustment and potentially mask the true effect of the primary exposure of interest. Following this suggestion, we have

removed traffic-related variables from our list of confounders in the final model to ensure the total effect is captured more accurately.

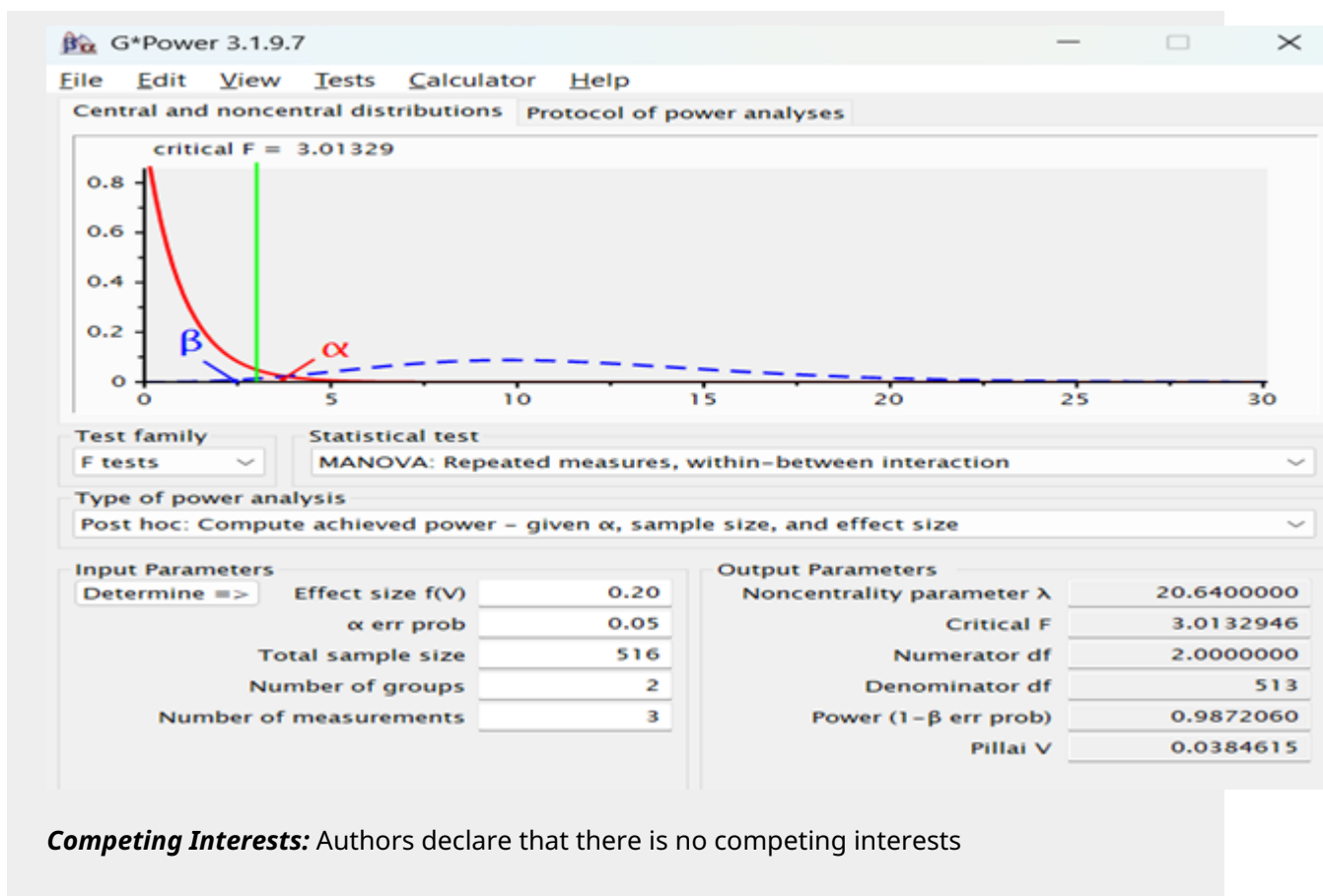
Thirdly, the monitors used has a statement that they have been used for research but that link is broken so impossible to assess how good they are. Low-cost sensors normally has to be validated against reference instruments. They also need to be adjusted for humidity in a non-linear way.

Response: We sincerely apologize for the broken link and the confusion regarding the sensor documentation. We have removed the inactive reference. To address the reviewer's concerns regarding sensor performance and environmental interference, we have added information about the validation measures we have done.

Validation against reference instrument: The sensors used in this study were locally calibrated against a collocated E-BAM (Beta Attenuation Monitor), a Federal Equivalent Method (FEM) standard instrument. We generated device-specific scatter plots which showed $R^2 > 0.88$. Following best practices, we derived and applied device-specific linear regression correction equations to all particulate matter(PM) data to align the low-cost sensor readings with the E-BAM standard. In addition to the E-BAM calibration, we applied a correction factor to account for hygroscopic growth of particles at high humidity levels, ensuring that the reported concentrations represent dry particle mass. We have updated the methods section to include specific calibration information and added scatter plots as a supplementary file to show the performance of the monitors against the standard instrument(Page No-12, Line no:526-541).

I am also a bit worried, as the other reviewer, about statistical power. If there are no big difference in exposure it might be even harder to find evidence.

Response: We acknowledge the reviewer's concern regarding statistical power. To address this, we conducted a power analysis using G*Power software. Based on our final sample size of 516 participants, the analysis confirms a statistical power of 98% (at an alpha level of 0.05) to detect effect size of $P = 0.20$. This indicates that the study is sufficiently powered to identify significant associations even if the differences in exposure levels are relatively small. The same has been updated in the method section.



Reviewer Report 06 July 2020

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Kalpna Balakrishnan

¹ Department of Environmental Health Engineering, Sri Ramachandra Institute of Higher Education and Research, Chennai, Tamil Nadu, India

² Department of Environmental Health Engineering, Sri Ramachandra Institute of Higher Education and Research, Chennai, Tamil Nadu, India

Please see my full review [here](#).

Competing Interests: No competing interests were disclosed.

I confirm that I have read this submission and believe that I have an appropriate level of

expertise to confirm that it is of an acceptable scientific standard, however I have significant reservations, as outlined above.

Version 2

Reviewer Report 22 January 2020

<https://doi.org/10.21956/wellcomeopenres.16960.r36890>

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**Ryan W. Allen**

¹ Simon Fraser University, Burnaby, BC, Canada

² Simon Fraser University, Burnaby, BC, Canada

³ Simon Fraser University, Burnaby, BC, Canada

I appreciate the opportunity to review and comment on version 2 of this paper. I have read the updated paper, the authors' responses to Dr. Kalpana Balakrishnan's comments on version 1, and Dr. Julian Marshall's comments on version 2.

As I indicated in my review of version 1, this is an important research area. I remain very enthusiastic about the idea of a birth cohort focused on prenatal air pollution exposure and these outcomes. However, the authors have not addressed several of my previous concerns:

- I previously commented that "The investigators plan to use the VAYUCARE device for measuring air pollutants, but the paper does not provide sufficient detail on this instrument or a rationale for choosing it. Has the device been validated against other, more established monitors?"

I remain concerned about the lack of validation data for the VAYUCARE. I agree with Dr. Marshall's recent comments that the VAYUCARE needs to be compared to a "gold standard" device and that six hours is insufficient for such a comparison.

- In my previous comments, I also wrote that "The authors should describe their plans for deriving exposure estimates from the measurements. For example, personal monitoring is planned during two 24-hour periods in pregnancy. These will need to be temporally adjusted and combined in some way to estimate long-term exposure during pregnancy."

The authors do not appear to have addressed this concern. It remains unclear how two 24-hour measurements will be used to estimate exposure over the full pregnancy or during specific trimesters. These short-term measurements will be heavily influenced by temporal variations in concentrations, and that variation will need to be accounted for in some way. Dr. Balakrishnan expressed a related concern in a comment about model specifications, but

after reading the authors' responses it is still unclear how this issue will be dealt with.

- I am also still concerned about the power calculations. In response to previous comments, the authors updated their calculations based on a systematic review of solid fuel use (Tielsch *et al.*, 2009). This is a surprising choice because, according to Dr. Balakrishnan, "the prevalence of solid fuel use is not likely to be higher than 5%." Moreover, a key driver of statistical power is the exposure gradient, but the power calculation still does not specify what gradient is being assumed.

Other comments:

- The paper indicates that "the recruitment and follow-up visits are scheduled between August 2018 and December 2020." Has data collection already started, or have the timelines not been updated?
- Will the authors have access to any data on gestational age at birth? For the primary outcome, LBW, gestational age is not needed. But it may also be useful to distinguish between children who were growth restricted vs. those who were born premature (for example, these may pose different risks for adiposity in childhood). For the former, an outcome like small for gestational age might be useful.
- Given emerging evidence that air pollution contributes to pregnancy loss, I encourage the authors to think through the potential role of "live birth bias" in this study (see citations Liew *et al.*, 2015¹ and Nobles and Hamoudi, 2019² for more details).

References

1. Liew Z, Olsen J, Cui X, Ritz B, et al.: Bias from conditioning on live birth in pregnancy cohorts: an illustration based on neurodevelopment in children after prenatal exposure to organic pollutants. *Int J Epidemiol.* 2015; **44** (1): 345-54 [PubMed Abstract](#) | [Publisher Full Text](#)
2. Nobles J, Hamoudi A: Detecting the Effects of Early-Life Exposures: Why Fecundity Matters. *Population Research and Policy Review.* 2019; **38** (6): 783-809 [Publisher Full Text](#)

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Air pollution exposure assessment and epidemiology.

I confirm that I have read this submission and believe that I have an appropriate level of expertise to state that I do not consider it to be of an acceptable scientific standard, for reasons outlined above.

Author Response 19 Feb 2020

Giridhara R Babu

1. I previously commented that "The investigators plan to use the VAYUCARE device for measuring air pollutants, but the paper does not provide sufficient detail on this instrument or a rationale for choosing it. Has the device been validated against other,

more established monitors?"

I remain concerned about the lack of validation data for the VAYUCARE. I agree with Dr. Marshall's recent comments that the VAYUCARE needs to be compared to a "gold standard" device and that six hours is insufficient for such a comparison.

Author: The reason behind selecting "VAYUCARE" for air pollution measurement in India, is that "Ambience monitoring India" is the only company in India that supplies user-friendly air quality devices which measures combined pollution parameters such as particulate matter (<10 µm, <2.5 µm) and carbon monoxide at the affordable price within our budgeted amount.

As we already mentioned, we have calibrated these low cost sensors against the - Karnataka State Pollution Control Board monitor. We appreciate and accept the invitation of Dr. Julian Marshall and are ready to validate our devices against BAM reference grade monitor.

2. In my previous comments, I also wrote that "The authors should describe their plans for deriving exposure estimates from the measurements. For example, personal monitoring is planned during two 24-hour periods in pregnancy. These will need to be temporally adjusted and combined in some way to estimate long-term exposure during pregnancy."

The authors do not appear to have addressed this concern. It remains unclear how two 24-hour measurements will be used to estimate exposure over the full pregnancy or during specific trimesters. These short-term measurements will be heavily influenced by temporal variations in concentrations, and that variation will need to be accounted for in some way. Dr. Balakrishnan expressed a related concern in a comment about model specifications, but after reading the authors' responses it is still unclear how this issue will be dealt with.

Author: Thank you for pointing out the issue. We agree that this is a practical limitation. Thus we are assuming that one measurement for 24 hours is a proxy for the exposure through that trimester. Since our funding limits the assessment throughout the trimester for now. We agree with your concern on temporal variability in estimates for that we have collected detailed information on 24-hour activity pattern and the season associated at the time of measurement. We consider the long term exposure measurements of all three parameters at the ambient level using data extracted from the state pollution control board (annual average for that year).

3. I am also still concerned about the power calculations. In response to previous comments, the authors updated their calculations based on a systematic review of solid fuel use (Tielsch et al., 2009). This is a surprising choice because, according to Dr. Balakrishnan, "the prevalence of solid fuel use is not likely to be higher than 5%." Moreover, a key driver of statistical power is the exposure gradient, but the power calculation still does not specify what gradient is being assumed.

Author: Thank you once again for raising this point, however, since we have already begun our data collection as per our planned timeline, we are unable to make any modification in

terms of sample size now. The current sample size is calculated using the outcome gradient which was found to be around 20% in Karnataka provided by Ramesh Chellan et al (Ref. No 44).

4. The paper indicates that "the recruitment and follow-up visits are scheduled between August 2018 and December 2020." Has data collection already started, or have the timelines not been updated?

Author: Yes, we have started the data collection and mentioned the timeline for follow up visit at birth and one year. The recruitment and follow up will go hand in hand.

5. Will the authors have access to any data on gestational age at birth? For the primary outcome, LBW, gestational age is not needed. But it may also be useful to distinguish between children who were growth restricted vs. those who were born premature (for example, these may pose different risks for adiposity in childhood). For the former, an outcome like small for gestational age might be useful.

Author: Many thanks for your suggestions. We do have access to information of gestational age at delivery by means of last menstrual period date and actual date of delivery and will try to look for the small for gestational age. Additionally as our eligibility criteria is inclusion of pregnant women before 18 weeks of gestation, we also intend to capture pregnant women in early trimester.

6. Given emerging evidence that air pollution contributes to pregnancy loss, I encourage the authors to think through the potential role of "live birth bias" in this study (see citations Liew et al., 2015¹ and Nobles and Hamoudi, 2019² for more details).

Author: Thank you very much for the suggestion. We have now included the "live birth bias" in the bias section of our paper.

Competing Interests: No competing interests

Reviewer Report 12 December 2019

<https://doi.org/10.21956/wellcomeopenres.16960.r36891>

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Julian D. Marshall

1

Department of Civil and Environmental Engineering, University of Washington, Seattle, WA, USA

² Department of Civil and Environmental Engineering, University of Washington, Seattle, WA, USA

³ Department of Civil and Environmental Engineering, University of Washington, Seattle, WA, USA

I have reviewed the revisions to:

Ambient and Indoor Air Pollution in Pregnancy and the risk of Low birth weight and Ensuing Effects in Infants (APPLE): A cohort study in Bangalore, South India

Shriyan P, Babu GR, Ravi D, Ana Y, van Schayck OC, Thankachan P and Murthy G.

To review these revisions, I read the following:

- The authors' 1-paragraph summary of changes.
- The authors' reply to comments by Dr. Balakrishnan.
- The parts of the new document that are relevant to the issue I focused on in my initial review: the low-cost sensors.
- I also have reviewed a relevant citation from the new document; see below.

As a reminder, the two main concerns I raised in my initial review are:

1. "I am concerned that the proposal's current plan for air pollution measurement is insufficient; the level of accuracy and precision in those measurements may be too low, and as a result, exposure misclassification may overwhelm the statistical signal the authors seek to quantify."
2. "I agree with the review comments from Professors Balakrishnan and Allen regarding air pollution measurement."

I remain concerned about those two points.

I see these new aspects in the report:

- The authors' reply to Dr. Balakrishnan states: "Field calibration: the State Pollution Control Board of Karnataka (KSPCB) possesses several outdoor samplers, with some located in the city of Bangalore, to measure ambient CO and PM2.5 levels. For our calibration, we used the outdoor sampler located at the Indira Gandhi Institute of Child Health Hospital, Bangalore (device name: Ecotech Monitoring Solutions, NIMHANS). The board very frequently calibrates this device. During 6 hours, we placed our outdoor sampler and three of our indoor samplers directly next to the board's sampler. After that, we calibrated our devices against the board's sampler based on the outdoor readings. Correction factors were developed for all our devices."

Six hours is entirely insufficient to test a sensor, develop a calibration curve, and understand how robust the sensor is. The low-cost sensor also needs to be well calibrated against a gold-standard reference device, such as a BAM. This would involve, before commencing the study, many days or weeks of co-location with multiple low-cost sensors and at least one reference sampler. It would also involve, during the study, constantly testing the devices to see how their performance might be changing. See below - I could potentially help facilitate this calibration, if the authors are

interested.

- The authors' report states: "The monitors are calibrated, tested, and then installed⁴⁰." That sentence cites this report: http://www.janhensen.nl/team/past/master/Vervoort_2018.pdf. However, the report itself states that the devices were, in that study, not calibrated (see top of page 10); they were not compared against a gold-standard measure. The report also states "The test showed that PM2.5 is measured quite similarly by all monitors, see Figure 2.3" (page 10). However, Figure 2.3 instead shows that the devices do not internally agree with each other. (Also, the test duration was under 2 days, so even if the devices did agree, the test is not long enough to provide robust evidence.)

In conclusion, my two concerns above are unchanged by the revisions I saw.

As a minor note, my prior-round of review comments pointed out a minor typo in the abstract. In the revised version I read, the typo remains uncorrected. This should be edited.

Lastly, I am aware of two BAMs currently operating in Bangalore. If the authors would like to calibrate their low-cost sensors against a BAM (a reference device), and to see how the devices compare, they can reach out to me to discuss logistics; I can potentially help facilitate access. On a related note: I will be in Bangalore early next month (January); if it is of interest, I would be happy to meet to talk further in person about testing and calibrating the low-cost sensors.

Competing Interests: No competing interests were disclosed.

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard, however I have significant reservations, as outlined above.

Author Response 19 Feb 2020

Giridhara R Babu

1. "I am concerned that the proposal's current plan for air pollution measurement is insufficient; the level of accuracy and precision in those measurements may be too low, and as a result, exposure misclassification may overwhelm the statistical signal the authors seek to quantify."

Author: Thank you for the comment. We agree with the same that the level of accuracy and precision might be low in low cost sensors but it is an initial step to measure the individual exposure level at the household setting rather than depending on proxy measures from the ambient level pollution parameters. Our study also attempts to show its feasibility of measuring air pollution exposure at the community level. We agree with the misclassification bias which is occurring due to the exposure data collected as one of our study limitations.

2. "I agree with the review comments from Professors Balakrishnan and Allen regarding air pollution measurement."
I remain concerned about those two points.

I see these new aspects in the report:

The authors' reply to Dr. Balakrishnan states: "Field calibration: the State Pollution Control Board of Karnataka (KSPCB) possesses several outdoor samplers, with some located in the city of Bangalore, to measure ambient CO and PM2.5 levels. For our calibration, we used the outdoor sampler located at the Indira Gandhi Institute of Child Health Hospital, Bangalore (device name: Ecotech Monitoring Solutions, NIMHANS). The board very frequently calibrates this device. During 6 hours, we placed our outdoor sampler and three of our indoor samplers directly next to the board's sampler. After that, we calibrated our devices against the board's sampler based on the outdoor readings. Correction factors were developed for all our devices."

Six hours is entirely insufficient to test a sensor, develop a calibration curve, and understand how robust the sensor is. The low-cost sensor also needs to be well calibrated against a gold-standard reference device, such as a BAM. This would involve, before commencing the study, many days or weeks of co-location with multiple low-cost sensors and at least one reference sampler. It would also involve, during the study, constantly testing the devices to see how their performance might be changing. See below - I could potentially help facilitate this calibration, if the authors are interested.

Author: We agree with your concern, and appreciate your invitation. We have already discussed with your team regarding this and we will initiate this process soon.

3. The authors' report states: "The monitors are calibrated, tested, and then installed⁴⁰." That sentence cites this report:

http://www.janhensen.nl/team/past/master/Vervoort_2018.pdf. However, the report itself states that the devices were, in that study, not calibrated (see top of page 10); they were not compared against a gold-standard measure. The report also states "The test showed that PM2.5 is measured quite similarly by all monitors, see Figure 2.3" (page 10). However, Figure 2.3 instead shows that the devices do not internally agree with each other. (Also, the test duration was under 2 days, so even if the devices did agree, the test is not long enough to provide robust evidence.)

Author: Thank you for the comments. We want to clarify that the devices were calibrated and tested by the research team before its use in field against Karnataka State Pollution Control Board monitors. We have cited the reference to convey the message that it has been used by research purpose prior to use by us. (Reference no. 46)

4. As a minor note, my prior-round of review comments pointed out a minor typo in the abstract. In the revised version I read, the typo remains uncorrected. This should be edited.

Author: Thank you for pointing out the error, we have now corrected the same.

5. Lastly, I am aware of two BAMs currently operating in Bangalore. If the authors would like to calibrate their low-cost sensors against a BAM (a reference device), and to see how the devices compare, they can reach out to me to discuss logistics; I can potentially help facilitate access. On a related note: I will be in Bangalore early next month (January); if it is of interest, I would be happy to meet to talk further in person about testing and calibrating the low-cost sensors.

Author: We have already discussed with C-STEP team members and will initiate the process of calibration soon.

Competing Interests: No competing interests

Version 1

Reviewer Report 04 July 2019

<https://doi.org/10.21956/wellcomeopenres.16159.r35763>

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Julian D. Marshall

¹ Department of Civil and Environmental Engineering, University of Washington, Seattle, WA, USA

² Department of Civil and Environmental Engineering, University of Washington, Seattle, WA, USA

³ Department of Civil and Environmental Engineering, University of Washington, Seattle, WA, USA

⁴ Department of Civil and Environmental Engineering, University of Washington, Seattle, WA, USA

Thank you for this opportunity to review the study protocol, "Ambient and Indoor Air Pollution in Pregnancy and the risk of Low birth weight and Ensuing Effects in Infants (APPLE): A cohort study in Bangalore, South India [version 1]". In addition, I have read two peer review reports included with the manuscript: one by Kalpana Balakrishnan (Balakrishnan, 2018¹) and one by Ryan Allen (Allen, 2019²).

The proposed study focuses on the impact of air pollution on adverse birth outcomes. Air pollution concentrations for PM2.5, PM10, and CO will be measured indoors and in nearby (within 5km) ambient air, using the VAYUCARE low-cost sensor; measurements as 24-hour averages will be conducted twice per subject: once each during the second and the third trimesters. Health parameters will be investigated for the mother and the infant, focusing on two outcomes: adiposity and birth weight.

The research topics investigated here are very important. A successful investigation would meaningfully add to the literature.

To my knowledge, the health measurements and the plans for epidemiological analyses are robust. (Those aspects are not my area of expertise.) Comments below focus on the air pollution measurements.

I am concerned that the proposal's current plan for air pollution measurement is insufficient; the level of accuracy and precision in those measurements may be too low, and as a result, exposure misclassification may overwhelm the statistical signal the authors seek to quantify.

I agree with the review comments from Professors Balakrishnan and Allen regarding air pollution measurement. I would not want to see the important work of this project move forward, and then later the authors find out that the study is unsuccessful or inconclusive solely because of uncertainties in air pollution measurement.

To help shore up this aspect of the investigation, I suggest the authors conduct two exercises: first, a series of pilot studies (e.g., first without people, then on 5 people, then on 20 people, and then on 100 people, before expanding to the full ~500-person cohort). During this pilot study, the researchers should investigate the robustness of the air pollution measurements for use in their study. Strong attention is needed for issues such as error, bias, device reliability/failure rate, reproducibility, within- and between-instrument variability, within- and between-participant variability, interference (e.g., dependency on relative humidity), general spatial and temporal variability, and the range of concentrations observed. This information will usefully inform the study design (e.g., which device to use, how many measurements to make). Running multiple versions of the same device side-by-side will inform reproducibility; running them next to a reference monitor will inform accuracy. Second, a review on devices used to measure air pollution exposures for epidemiology, with special attention to issues of using low-cost, unproven sensors and to approaches that have been used in India.

The research would benefit from greater knowledge of expected concentrations and concentration gradients; such information would inform the sample-size/power calculations. Earlier measurements in Bangalore reported relatively high concentrations (e.g., Both *et al.*, 2011³); I believe current measurements in Bangalore by ILK and CSTEP, using a reference-grade monitor (Beta Attenuation Monitor [BAM]), suggest cleaner concentrations.

Minor typo:

The abstract contains the text "We will estimate the level of air pollutants including coarse particulate matter 10 ug/m³ (PM₁₀), fine particulate matter 2.5 ug/m³(PM_{2.5}) and carbon monoxide (CO) parts per million (ppm) levels in both indoor and ambient environment." Both uses of the phrase "ug/m³" should be "µm" (micrometers), reflecting the size of the particles (<10 µm, <2.5 µm) rather than a concentration measurement (mass per volume: µg/m³).

References

1. Balakrishnan K: Referee Report For: Ambient and Indoor Air Pollution in Pregnancy and the risk of Low birth weight and Ensuing Effects in Infants (APPLE): A cohort study in Bangalore, South India [version 1]. *Wellcome Open Research*. 2018; **3** (133). [Reference Source](#)
2. Allen RW: Referee Report For: Ambient and Indoor Air Pollution in Pregnancy and the risk of Low birth weight and Ensuing Effects in Infants (APPLE): A cohort study in Bangalore, South India [version 1]. *Wellcome Open Research*. 2019; **3** (133). [Reference Source](#)
3. Both AF, Balakrishnan A, Joseph B, Marshall JD: Spatiotemporal aspects of real-time PM(2.5):

low- and middle-income neighborhoods in Bangalore, India. *Environ Sci Technol*. 2011; **45** (13): 5629-36 [PubMed Abstract](#) | [Publisher Full Text](#)

Is the rationale for, and objectives of, the study clearly described?

Yes

Is the study design appropriate for the research question?

Partly

Are sufficient details of the methods provided to allow replication by others?

Yes

Are the datasets clearly presented in a useable and accessible format?

Not applicable

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Air pollution monitoring; exposure assessment

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard, however I have significant reservations, as outlined above.

Reviewer Report 06 June 2019

<https://doi.org/10.21956/wellcomeopenres.16159.r35547>

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Ryan W. Allen

¹ Simon Fraser University, Burnaby, BC, Canada

² Simon Fraser University, Burnaby, BC, Canada

³ Simon Fraser University, Burnaby, BC, Canada

⁴ Simon Fraser University, Burnaby, BC, Canada

This paper describes a birth cohort study to evaluate associations between air pollution exposure and low birth weight (LBW) in areas near Bangalore. The investigators also intend to evaluate the mediating role of LBW in associations between prenatal air pollution exposure and adiposity in early childhood. This is an ambitious study focused on important areas of inquiry. The scientific evidence linking air pollution with fetal growth is expanding rapidly, while the research on prenatal exposure and adiposity/overweight/obesity is relatively limited and results are mixed.

I have several concerns with the study as described:

1. Exposure assessment:

- The measurement plan is difficult to understand, due in part to inconsistent use of terminology. Under “exposure assessment”, the authors describe plans for outdoor, indoor, and personal sampling. Then under “Data sources and measurements” the authors describe only ambient and indoor monitoring, with no mention of personal monitoring. The paper states that “personal sampling monitors will be used to measure...individual exposure...at the household level.” Are the terms “personal” and “indoor” being used interchangeably? Or will each participant undergo both personal and indoor sampling? If so, will these be conducted at the same time?
- The investigators plan to use the VAYUCARE device for measuring air pollutants, but the paper does not provide sufficient detail on this instrument or a rationale for choosing it. Has the device been validated against other, more established monitors? I recommend that the authors carefully consider their choice and provide a clear rationale for choosing this particular instrument. As interest in relatively low-cost monitors has increased, there have been more evaluations of various monitors and their strengths and weaknesses, and the authors might consider reviewing some of these recent evaluations such as:
 - <http://www.aqmd.gov/aq-spec/evaluations>
 - <https://www.epa.gov/air-sensor-toolbox/evaluation-emerging-air-pollution-sensor-performance>
 - Morawska *et al.*, 2018¹.
- The authors should describe their plans for deriving exposure estimates from the measurements. For example, personal monitoring is planned during two 24-hour periods in pregnancy. These will need to be temporally adjusted and combined in some way to estimate long-term exposure during pregnancy. In addition, it is not clear if the investigators have plans to estimate exposure prior to enrolment in the study (the period from conception to enrolment).
- The investigators plan to place outdoor monitors within 5 km of participants’ residences. This may be too far to assess concentrations at the residence location. I suggest that the team consider models for interpolating outdoor measurements.

2. Power calculations:

- The study’s power to detect associations will depend on the exposure gradient, but it is not clear what gradient the investigators assumed for their power calculations. Moreover, the ambiguity in the exposure assessment makes the power calculation difficult to interpret; it is unclear whether the exposure of interest is the pregnancy averaged concentration, trimester-specific concentrations, or concentrations over some other averaging time.
- The investigators note the transient nature of this population in assuming a loss of 40% of the cohort. It is not clear if the investigators also accounted for pregnancy losses in their anticipated loss to follow-up. One might expect that $\geq 10\%$ of these pregnancies will result in spontaneous abortion or stillbirth (depending on gestational age at enrolment).

3. Analysis plan:

- Hypotheses 3 and 4 assume an association between air pollution during pregnancy and adiposity in childhood. But the literature on this question is mixed, so it is possible that air pollution will not be associated with adiposity in this cohort. I recommend that the authors revise their hypotheses to formally test the air pollution – adiposity association before moving to the mediation analysis.

- For continuous outcome variables in childhood, will each time point be modelled separately or are there plans to take advantage of the study's longitudinal design to evaluate growth trajectories?
- The literature suggests that rapid "catch-up" growth after birth is strongly tied to poor cardiometabolic health later in life. Thus, in addition to the mediation analysis of LBW on the exposure-adiposity pathway, I suggest that the authors consider also explicitly evaluating catch-up growth as an outcome.
- The paper does not describe how the biomarkers (haemoglobin, oral glucose) will be used. These variables are not listed as potential confounders or possible effect modifiers (in Figure 3 or Table 1), nor are they outcome variables. Collection of these samples will pose a burden to participants, so the use of these samples needs to be justified.

In addition, I have the following minor comments/suggestions:

- The summary of previous studies on air pollution and fetal growth could be updated and expanded to include, for example, some of the papers listed at the end of this review.
- Figure 6 would be more informative as a timeline showing key activities at different stages of pregnancy/childhood.
- The questionnaire (supplementary file 1) needs refinement. Some questions are vague/subjective and may not provide useful information (e.g. "Is that a dusty occupation?", "Please describe the weather").

References

1. Morawska L, Thai PK, Liu X, Asumadu-Sakyi A, et al.: Applications of low-cost sensing technologies for air quality monitoring and exposure assessment: How far have they gone?. *Environ Int.* **116**: 286-299 [PubMed Abstract](#) | [Publisher Full Text](#)
2. Jacobs M, Zhang G, Chen S, Mullins B, et al.: The association between ambient air pollution and selected adverse pregnancy outcomes in China: A systematic review. *Sci Total Environ.* 2017; **579**: 1179-1192 [PubMed Abstract](#) | [Publisher Full Text](#)
3. Li X, Huang S, Jiao A, Yang X, et al.: Association between ambient fine particulate matter and preterm birth or term low birth weight: An updated systematic review and meta-analysis. *Environ Pollut.* 2017; **227**: 596-605 [PubMed Abstract](#) | [Publisher Full Text](#)
4. Fu L, Chen Y, Yang X, Yang Z, et al.: The associations of air pollution exposure during pregnancy with fetal growth and anthropometric measurements at birth: a systematic review and meta-analysis. *Environ Sci Pollut Res Int.* 2019. [PubMed Abstract](#) | [Publisher Full Text](#)
5. Klepac P, Locatelli I, Korošec S, Künzli N, et al.: Ambient air pollution and pregnancy outcomes: A comprehensive review and identification of environmental public health challenges. *Environ Res.* **167**: 144-159 [PubMed Abstract](#) | [Publisher Full Text](#)
6. Melody SM, Ford J, Wills K, Venn A, et al.: Maternal exposure to short-to medium-term outdoor air pollution and obstetric and neonatal outcomes: A systematic review. *Environ Pollut.* 2019; **244**: 915-925 [PubMed Abstract](#) | [Publisher Full Text](#)
7. Yuan L, Zhang Y, Gao Y, Tian Y: Maternal fine particulate matter (PM_{2.5}) exposure and adverse birth outcomes: an updated systematic review based on cohort studies. *Environ Sci Pollut Res Int.* 2019; **26** (14): 13963-13983 [PubMed Abstract](#) | [Publisher Full Text](#)

Is the rationale for, and objectives of, the study clearly described?

Partly

Is the study design appropriate for the research question?

Partly

Are sufficient details of the methods provided to allow replication by others?

No

Are the datasets clearly presented in a useable and accessible format?

Not applicable

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Air pollution exposure assessment and epidemiology.

I confirm that I have read this submission and believe that I have an appropriate level of expertise to state that I do not consider it to be of an acceptable scientific standard, for reasons outlined above.

Reviewer Report 13 May 2019

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Kalpana Balakrishnan 

¹ Department of Environmental Health Engineering, Sri Ramachandra Institute of Higher Education and Research, Chennai, Tamil Nadu, India

² Department of Environmental Health Engineering, Sri Ramachandra Institute of Higher Education and Research, Chennai, Tamil Nadu, India

³ Department of Environmental Health Engineering, Sri Ramachandra Institute of Higher Education and Research, Chennai, Tamil Nadu, India

⁴ Department of Environmental Health Engineering, Sri Ramachandra Institute of Higher Education and Research, Chennai, Tamil Nadu, India

The proposal focuses on a theme that is being increasingly recognized as an important health outcome associated with air pollution exposures. This is especially pertinent in low-middle income countries where the prevalence of health damaging exposures and low birth weight are both high. The investigators have previously been engaged with cohort studies and thus have experience in launching such studies. The health outcomes section of the proposal is quite strong.

The proposal unfortunately does not provide a correct framing for addressing air pollution exposures and uses terminologies rather loosely making it difficult to interpret the rationale, objectives and the study methods proposed. A detailed listing of the issues and suggestions are provided below for possible consideration by the investigators and the funders

prior to a final decision:

1. The term "indoor air pollution" is not defined correctly. The investigators seem to want to address air pollution exposures from solid fuel use and term it as indoor air pollution. Air pollution resulting from solid fuel combustion is defined by WHO as household air pollution and contributes to both indoor and ambient air pollution. However in the study population chosen (urban slums near Bangalore) the contributions from solid fuel use to indoor and ambient air pollution is likely to be negligible. In the event it is not so, the investigators should provide supporting information to make a case for prevalence of solid fuel use in the study population. The all India average of 67% solid fuel use is dated and currently stands at 59%. In large metropolitan cities such as Bangalore, even in slums the prevalence of solid fuel use is not likely to be greater than 5%. Indoor air pollution in this setting therefore will not represent contributions from solid fuel use.

Ambient air pollution in the setting proposed (based on recent source apportionment studies) is primarily from transport-related emissions and is likely to be responsible for bulk of the indoor exposures, especially in slums. In the absence of other prominent indoor sources (as is likely in urban slums) ambient/indoor ratios for most pollutants will be close to 1.

The study design and methods thus cannot examine the independent association of indoor vs ambient air pollution with birth weight in this setting, nor can it address household air pollution from solid fuel use.

2. The investigators have not used the most recent or pertinent literature for the proposal. See several citations that have been added that provide a more comprehensive background on the topic. In particular recent studies in Chennai have documented the impacts on birthweight in a rural-urban cohort that address the dual risk of household and ambient air pollution through longitudinal household area measurements. A similar such study (DAPHNE) is underway in Delhi that focuses on personal exposures for urban pregnant women.
3. The sample size is not well informed by the expected exposure variability in the setting. The average exposure used in the cited reference (Balakrishnan *et al.*, 2018¹) is a national average for ambient air pollution. This does not address expected variability in personal exposures in an urban slum. It is also not clear if the effects estimate they have used is informed by the recent Chennai cohort study or meta-analytical estimates from other LMIC studies. Further performing two measurements may not adequately capture temporal variability (assuming that indeed women are able to consistently participate in a third trimester personal measurement). A 10000mAh power bank can be quite cumbersome to carry and may pose some safety/personal concerns for pregnant women.
4. The investigators do not provide any validation/citations/specifications for the instrument to be used for personal exposure monitoring. It is not clear if this is a new generation low cost sensor and if so, appropriate validations against reference monitors would be needed to

develop quantitative exposure-response relationships. CO exposure is unlikely to be of concern in an urban population (given the expected low prevalence of solid fuel use). CO in the ambient environment is unlikely to be detected at good enough resolution in low cost sensors.

5. Numerous infant outcomes are being measured including growth and development milestones for which there exists a stronger pool of evidence for air pollution-related impacts but yet only mediations through pre-natal exposures on adiposity is being examined. It would be immensely useful to measure post-birth so that each of the outcomes could be independently studied in addition to mediation effects. The sample size requirement will however likely be much higher.
6. Finally, the model specifications need to be elaborated further. Given the spatio-temporal variability and the inability to characterize this fully with just two measurements during pregnancy, there will presumably be a need for mixed effects models to estimate long-term (pregnancy period) exposures. Planned sensitivity analyses should be described in greater detail.

The exposure assessment components of the proposal need to be totally re-structured to focus only on ambient air pollution, using alternative exposure measurement approaches, in a larger cohort and using well validated instrumentation. A wealth of high quality health information is being collected but without a similar rigour in exposure measurements, the ability to develop quantitative exposure-response would not be realized.

Much has advanced in terms of using alternative exposure assessment methodologies in large cohorts. These should find a place in this cohort as well.

References

1. Balakrishnan K, Ghosh S, Thangavel G, Sambandam S, et al.: Exposures to fine particulate matter (PM_{2.5}) and birthweight in a rural-urban, mother-child cohort in Tamil Nadu, India. *Environ Res.* **161**: 524-531 [PubMed Abstract](#) | [Publisher Full Text](#)
2. Jedrychowski W, Perera F, Mrozek-Budzyn D, Mroz E, et al.: Gender differences in fetal growth of newborns exposed prenatally to airborne fine particulate matter. *Environ Res.* 2009; **109** (4): 447-56 [PubMed Abstract](#) | [Publisher Full Text](#)
3. Sun X, Luo X, Zhao C, Zhang B, et al.: The associations between birth weight and exposure to fine particulate matter (PM_{2.5}) and its chemical constituents during pregnancy: A meta-analysis. *Environ Pollut.* 2016; **211**: 38-47 [PubMed Abstract](#) | [Publisher Full Text](#)
4. Arku RE, Birch A, Shupler M, Yusuf S, et al.: Characterizing exposure to household air pollution within the Prospective Urban Rural Epidemiology (PURE) study. *Environ Int.* **114**: 307-317 [PubMed Abstract](#) | [Publisher Full Text](#)

Is the rationale for, and objectives of, the study clearly described?

No

Is the study design appropriate for the research question?

Partly

Are sufficient details of the methods provided to allow replication by others?

Yes

Are the datasets clearly presented in a useable and accessible format?

Not applicable

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Exposure assessment, air pollution and health effects, environmental epidemiology, bio-monitoring, environmental policy

I confirm that I have read this submission and believe that I have an appropriate level of expertise to state that I do not consider it to be of an acceptable scientific standard, for reasons outlined above.

Author Response 01 Oct 2019

Giridhara R Babu

The proposal focuses on a theme that is being increasingly recognized as an important health outcome associated with air pollution exposures. This is especially pertinent in low-middle income countries where the prevalence of health damaging exposures and low birth weight are both high. The investigators have previously been engaged with cohort studies and thus have experience in launching such studies. The health outcomes section of the proposal is quite strong.

Many thanks for recognizing the strengths of the proposal.

The proposal, unfortunately, does not provide a correct framing for addressing air pollution exposures and uses terminologies rather loosely making it difficult to interpret the rationale, objectives and the study methods proposed. A detailed listing of the issues and suggestions are provided below for possible consideration by the investigators and the funders prior to a final decision:

Our sincere thanks to your time and effort in providing detailed comments. We respect each of the inputs that you have given as a well-established senior mentor and researcher in this field. We have answered your queries in multiple parts to provide valid reasons, rationale, and details on methods. However, we earnestly submit that an appeal to the funders in a platform for peer review of the protocol is unwarranted.

1. The term "indoor air pollution" is not defined correctly. The investigators seem to want to address air pollution exposures from solid fuel use and term it as indoor air pollution. Air pollution resulting from solid fuel combustion is defined by WHO as household air pollution and contributes to both indoor and ambient air pollution. However, in the study population chosen (urban slums near Bangalore) the contributions from solid fuel use to indoor and ambient air pollution is likely to be negligible. In the event it is not so, the investigators should provide supporting information to make a case for prevalence of solid fuel use in the study population. The all India average of 67% solid fuel use is dated and currently stands at 59%. In large metropolitan cities such as Bangalore, even in slums, the prevalence of solid fuel use is not likely to be higher than 5%. Indoor air pollution in this setting, therefore, will not represent contributions from solid fuel use.

Ambient air pollution in the setting proposed (based on recent source apportionment studies) is primarily from transport-related emissions and is likely to be responsible for bulk of the indoor exposures, especially in slums. In the absence of other prominent indoor sources (as is likely in urban slums) ambient/indoor ratios for most pollutants will be close to 1.

The study design and methods thus cannot examine the independent association of indoor vs ambient air pollution with birth weight in this setting, nor can it address household air pollution from solid fuel use.

Thank you for your valuable inputs on this. We have not defined the solid fuel as indoor air pollution in our study. Our objective is to see the total effect of individual's exposure to air pollutants mainly PM2.5, PM10, and CO inside the house as well as their levels in the ambient air during the during the second and third trimester of pregnancy. The exposure of interest in our study is the levels of PM2.5, PM10, and CO, not solid fuel use. Solid fuel use is a contributing factor for these pollutants. Not only cooking fuel but several other contributing factors like incense sticks, frankincense, mosquito repellents (solid/liquid) are also leading to a higher amount of exposure. Your study has provided a detailed note of all these sources. We have obtained information regarding all these sources and not just solid fuels. However, we are treating only the pollutants (as a result of all these sources) as exposure. Pregnant women are vulnerable to inhalation of a higher level of pollutants , due to the poor housing structure and inadequate ventilation in the urban slums. Our focus is not only on solid fuel use, there are several other activities done at the household level which contribute to the emission of higher level of PM2.5, PM10 and CO. We are recording each of these activities at the household occurring at regular basis. Prospective assessment at early pregnancy is the best design to prove the independent association between exposure to pollution and birth outcomes. We are recording the level of personal exposure to air pollutants during pregnancy, prospectively following them up until delivery to see the effect on birth weight of the baby. We are not here to address the air pollution from solid fuel use. Our findings may provide the contributing factors for higher level of PM2.5, PM10, and CO at the individual level. It is unfortunate that most of your comments are based on the incorrect assumption that we are assessing only solid fuel use. We request you to please see the questionnaire and be assured that this is not what we are doing.

We agree that the study design and methods cannot examine the independent association of indoor vs. ambient air pollution, but it is designed to examine the total effect of all pollutants (irrespective of indoor v/s outdoor) with birth weight in this setting. Accordingly, we have made changes in the manuscript. We did not intend to address estimate household air pollution from solid fuel use.

1. The investigators have not used the most recent or pertinent literature for the proposal. See several citations that have been added that provide a more comprehensive background on the topic. In particular recent studies in Chennai have documented the impacts on birth weight in a rural-urban cohort that address the dual risk of household and ambient air pollution through longitudinal household area measurements. A similar such study (DAPHNE) is underway in Delhi that focuses on personal exposures for urban pregnant women.

Thank you for the comments. The recent literature has been updated in the Background section. The recent articles have been quoted in page no. 3, line no 17 and 24.

1. The sample size is not well informed by the expected exposure variability in the setting. The average exposure used in the cited reference (Balakrishnan *et al.*, 2018¹) is a national average for ambient air pollution. This does not address expected variability in personal exposures in an urban slum. It is also not clear if the effects estimate they have used is informed by the recent Chennai cohort study or meta-analytical estimates from other LMIC studies. Further performing two measurements may not adequately capture temporal variability (assuming that indeed women are able to participate in a third trimester personal measurement consistently). A 10000mAh power bank can be quite cumbersome to carry and may pose some safety/personal concerns for pregnant women.

Thank you for the comment. We have now re-calculated the sample size using a published systematic review report/ risk estimates reported by Tielsch et al, 2009. Risk of Low Birth Weight and Stillbirth Associated With Indoor Air Pollution From Solid Fuel Use in Developing Countries-2010: Increased risk of LBW of 38% (OR ¼ 1.38, 95% CI: 1.25, 1.52) from exposure to IAP and an associated reduction in birth weight of 96.6 g[1] Tielsch et al, 2009: Adjusted analysis: Risk of Low birth weight by exposure to solid biomass fuel [RR (term) ¼ 1.49, 95% CI: 1.25, 1.77; RR (preterm) ¼ 1.70, 95% CI: 0.93, 3.10] A cohort study][2]

We agree with you and have planned our measurements trimester-wise, but through field experiences, we have noticed that the majority of the study population contact the health center after completion of the first trimester. This is a practical limitation. We are assuming that one measurement for 24 hours is a proxy for the exposure through that trimester. The funding available does not permit us to assess throughout the trimester for now.

The device weighs approximately 110gms, and the power bank is approximately 200gm at the most. We have designed a sling bag to hold the device along with a power bank

and asked specific questions to the study subjects on any discomfort they felt wearing devices. In our initial assessment of around 100 women, except for two women, rest all did not report any discomfort while wearing the bag. The reason for discomfort by these two women was that it is a disturbance in carrying the bag while doing household chores.

1. The investigators do not provide any validation/citations/specifications for the instrument to be used for personal exposure monitoring. It is not clear if this is a new generation low-cost sensor and if so, appropriate validations against reference monitors would be needed to develop quantitative exposure-response relationships. CO exposure is unlikely to be of concern in an urban population (given the expected low prevalence of solid fuel use). CO in the ambient environment is unlikely to be detected at good enough resolution in low cost sensors.

Thanks for the concerns. Please note that a manuscript describing validation is under the process of publication elsewhere. The device we are using is a low cost monitoring device which measures parameters PM2.5, PM10, and CO. The manufacturer is Ambience Monitoring India, New Delhi. It is a custom-built device. The scattered light is transformed into signals, that are amplified and processed. The specifications are as follows:- Range of measuring parameter, Particulate Matter 2.5 (PM2.5) Fine particulate Range: 0.0 – 999.9 $\mu\text{g}/\text{m}^3$, Particulate Matter 10 (PM10) Course particulate Range: 0.0 -1999.9 $\mu\text{g}/\text{m}^3$ Minimum resolution of $< 0.3 \mu\text{m}$. Relative error Maximum of 15% and $\pm 10 \mu\text{g}/\text{m}^3$ Operating Temperature Range -10 degrees C to +50 degrees C Operating Humidity Maximum Limit 70% RH . Carbon Monoxide sensor: Type: Electrochemical Measurement Range: 0 - 1000 ppm Detection Limit: 0.5 ppm Resolution: $< 100 \text{ ppb}$ (instrumentation dependent) Precision: $< \pm 2 \%$ of reading Response: between 15 - 30 seconds Sensitivity: $4.75 \pm 2.75 \text{ nA/ppm}$ Expected operating life: > 5 years (up to 10 years if average temperature is $23^\circ\text{C} \pm 3$ degrees) Operating temperature range: -30 to 55°C (-20 to $+40^\circ\text{C}$ recommended) . Temperature: Range: -40 – 123°C Accuracy: $\pm 0.4^\circ\text{C}$ Power: 5uW standby, 3mw active Response: 8s (up to 20 minutes to balance if moved out of a location of excessive heat $35^\circ\text{C}+$). Relative Humidity: Range: 0 – 100% humidity Accuracy: $\pm 3\%$; $\pm 4\%$ up to 80% RH $\pm 5\%$ up to 90% RH Power: 5uW standby, 3mw active. Certification CE/FCC/RoHS.

The device has been validated before use. Initially, Zero PM2.5 and CO for both indoor and outdoor samplers performed using a sealed glass container $45\text{cm} \times 45\text{cm} \times 35\text{cm}$ with an inlet and outlet. The glass chamber contains two valves, one at the inlet and outlet that were closed during the zero-calibration measurements. All sides of the glass container were glued together to make it airtight, except for the lid. For the zero-calibration, we had placed our fully charged indoor and outdoor devices in the glass chamber after switching them on. After that, the lid was closed and sealed with double-sided tape. There was no gas inflow or outflow possible during the experiment. The devices were kept in the closed chamber until all their battery lives have ended (min 24h). The devices were kept in an airtight container without any exposure to CO or PM2.5.

Apart from the zero-calibration, also a one-point CO calibration has been performed in the same glass chamber. For this experiment, we have used calibrated gas cylinders of CO with a volume capacity of 10 litres with 50 ppm of CO concentration. After

switching the devices on, they were kept in ambient air for a period of minimum one hour so that the readings of the device stabilized in ambient air. After that, the device was placed in the glass container after which the lid was closed and sealed. At this point, the valve at the outlet was left open and at the same time, the CO gas from the cylinder was released through the inlet to force the present air out of the container. To ensure there was no mixture of gases remaining in the chamber to avoid contamination in the device readings, we connected the outlet pipe to a water container having a measuring cylinder and flushed CO double the volume of the glass chamber. The outlet valve was then closed, and the chamber at this point was filled with 50 ppm. The regulator was closed and the device was left inside the sealed container for an hour. The device was removed after that and left for another hour in ambient air to stabilize the readings post-exposure to CO.

Field calibration: the State Pollution Control Board of Karnataka (KSPCB) possesses several outdoor samplers, with some located in the city of Bangalore, to measure ambient CO and PM2.5 levels. For our calibration, we used the outdoor sampler located at the Indira Gandhi Institute of Child Health Hospital, Bangalore (device name: Ecotech Monitoring Solutions, NIMHANS). The board very frequently calibrates this device. During 6 hours, we placed our outdoor sampler and three of our indoor samplers directly next to the board's sampler. After that, we calibrated our devices against the board's sampler based on the outdoor readings. Correction factors were developed for all our devices.

1. Numerous infant outcomes are being measured including growth and development milestones for which there exists a stronger pool of evidence for air pollution-related impacts but yet only mediations through pre-natal exposures on adiposity is being examined. It would be immensely useful to measure post-birth so that each of the outcomes could be independently studied in addition to mediation effects. The sample size requirement will however likely be much higher.

We agree with you. The current study is a subset of the larger cohort ongoing in Bangalore Government Health facilities with a targeted sample size of 5000 pregnant women with an aim to understand the transgenerational role of glucose and other nutrients, psychosocial environment on the poor infant outcomes and risk of childhood obesity, as an early marker of chronic diseases. Through the recent evidences it has been also proved that air pollution is also a contributing factor for poor infant outcomes. Hence we built our research question on evaluating independent association between indoor and ambient level of PM2.5, PM10 and CO on birth outcomes. We are in constant search for funding. In order to measure several post-birth outcomes independently studied in addition to mediation effects, we are seeking further funding from other sources as you have rightly pointed out that the sample size requirement will be much higher. We got funding for only four air quality monitoring devices and therefore, it has been difficult to measure the pollution parameters after birth. We will try to cover the post-birth measurements, through additional sources of funding.

1. Finally, the model specifications need to be elaborated further. Given the spatio-temporal variability and the inability to characterize this fully with just two

measurements during pregnancy, there will presumably be a need for mixed effects models to estimate long-term (pregnancy period) exposures. Planned sensitivity analyses should be described in greater detail.

The exposure assessment components of the proposal need to be restructured to focus only on ambient air pollution, using alternative exposure measurement approaches, in a larger cohort and using well-validated instrumentation. A wealth of high quality health information is being collected but without a similar rigour in exposure measurements, the ability to develop quantitative exposure-response would not be realized.

Much has advanced in terms of using alternative exposure assessment methodologies in large cohorts. These should find a place in this cohort as well.

We are using a multivariate linear regression model to examine the association between birth weight and pollution parameters (in continuous variables) separately adjusting for listed potential confounders. From the apriori, information covariates will be selected. Forming the causal path from exposure to outcome potential confounders will be listed. The model will be developed by adjusting to various potential confounders. A model with larger adjusted R2 and predicted R2 value would be considered and also with all confounders, we will run step-wise regression analysis. We will also categorize the exposure variables trimester wise as per the percentiles and run multivariate model adjusting for known important potential confounder variables. As we progress, we are keen to included advanced techniques using alternative exposure assessment methodologies in large cohorts. Your advice in this regard is of high value.

Again, we sincerely thank you for all the comments.

Competing Interests: No competing interests were disclosed.

Comments on this article

Version 3

Author Response 28 Aug 2026

Giridhara R Babu

RESPONSE TO REVIEWERS Ambient and Indoor Air Pollution in Pregnancy and the risk of Low birth weight and Ensuing Effects in Infants (APPLE): A cohort study in Bangalore, South India

Article type: Study Protocol · **Journal:** Wellcome Open Research · **Article:** 2018, 3:133 ·

Revision: version 4

We thank all seven reviewers for the time and care they have given to this protocol across four versions. The comments on version 3 from Professor Malmqvist, Professor Lu, Dr Liao, and Dr Castelhanos led to substantive changes to the exposure assessment, calibration, power, and analysis sections, and we are grateful for them. We have also addressed the outstanding points raised by Professor Balakrishnan, Professor Allen, and Professor Marshall on earlier versions, as Dr Liao notes that he read those reports; we address these in a separate section at the end of this document. Reviewer comments are reproduced in shaded boxes. Our response follows each comment, with the location of the corresponding change in the revised manuscript given in bold. Page and line references are to the clean revised manuscript accompanying this response.

Summary of the principal changes in version 4 **Area What changed** Exposure assessment New subsection describing how two 24-hour personal measurements are temporally adjusted against the continuous ambient record to derive trimester-specific and pregnancy-average exposures, with ICC testing, regression calibration and an explicit statement of the expected direction of bias. Sensor calibration New subsection on calibration against a co-located E-BAM reference instrument ($R^2 > 0.88$), device-specific correction equations, humidity correction for hygroscopic growth, and periodic re-checks. The broken manufacturer link has been removed. Calibration scatter plots added as Supplementary Figures 1 and 2. Confounder set Traffic and roadway proximity removed from the confounder set, with the over-adjustment rationale stated explicitly. Traffic retained as a determinant of exposure and a candidate effect modifier. Statistical power Post-hoc power replaced with a minimum detectable effect analysis reported against defined exposure contrasts, together with an explicit statement about the exposure gradient. Housing and cooking covariates Expanded description of the ventilation, housing, cooking-location, fuel, cook identity, cooking duration and time-activity data already collected. Adiposity mechanism New Background paragraph setting out the placental, inflammatory, adipokine and catch-up-growth pathways linking prenatal air pollution to infant adiposity. Analysis plan Planned sensitivity analyses stratified by personal and household factors added, together with sensitivity analyses for exposure windows, measurement error, humidity correction and live-birth bias. Laboratory section Role of MEDALL under the memorandum of understanding clarified; timing of blood collection specified as at or after 24 weeks of gestation. Figure 2 Map redrawn with a labelled lat/long graticule, north arrow, scale bar, legend, stated projection and base map attribution on both panels; panels labelled A and B; facility label spelling corrected. Notation PM_{10} ($\leq 10 \mu m$) and $PM_{2.5}$ ($\leq 2.5 \mu m$) notation corrected throughout the abstract and manuscript. References Eleven references added (53–63). Existing numbering 1–52 is unchanged. Discussion Strengths, limitations and an environmental justice / future work paragraph added.

Reviewer 4: Professor Ebba Malmqvist, Lund University, Sweden We thank Professor Malmqvist for a set of comments that go to the heart of the exposure assessment, and which have produced the largest changes in this version.

Comment 4.1 *My first caveat is to assess exposure by 2 times 24 hour measurements. I think this needs to be complemented by questionnaires both in general and during the exposure assessment. This should include questions on ventilation (number of rooms, doors and windows) and if cooking is done outside, in the same room as sleeping or in a separate room. Type of fuel used and time for cooking and who performs the task.* We agree that behavioral and environmental context is essential for interpreting short-term exposure measurements. We collected these data, but the previous version described them too briefly for the reviewer to see this, and we have now set them out in full. Two instruments are used. A general questionnaire administered at recruitment records the number of rooms in the

dwelling, the number of external doors and windows, the presence and type of kitchen ventilation (window, exhaust fan, chimney or none), whether windows are kept open during cooking, the location of cooking (outdoors, in a separate kitchen, or in the room also used for sleeping), the primary and secondary fuels used, the identity of the primary cook, the number of meals cooked per day and the total duration of cooking. A post-monitoring questionnaire, administered when the sampler is retrieved, records the participant's time-activity pattern over the monitored 24 hours, the meals and food types prepared during that period, and any unusual combustion events. Secondary indoor sources — incense, mosquito coils and repellents, candles — and tobacco use by household members are recorded in both instruments.

Change made: New paragraph "Housing, ventilation and cooking practices" added under **Covariates** (Methods).

Comment 4.2 *Secondly, I am a bit worried to see traffic as a confounder because it is part of the exposure and adjusting for it will remove effects possibly.* We accept this point without reservation. Traffic is a major source of the ambient particulate exposure that constitutes our exposure of interest, not a common cause of exposure and outcome. Conditioning on it would be over-adjustment and would remove part of the total effect we intend to estimate. We therefore removed traffic and roadway proximity from the confounder set. We have retained traffic proximity as a descriptive determinant of measured exposure and as a candidate effect modifier in a secondary analysis, which does not carry the same risk.

Change made: The **Confounders** paragraph under Covariates has been rewritten; traffic and roadways are removed from the adjustment set and the over-adjustment rationale is stated explicitly in the manuscript.

Comment 4.3 *Thirdly, the monitors used has a statement that they have been used for research but that link is broken so impossible to assess how good they are. Low-cost sensors normally has to be validated against reference instruments. They also need to be adjusted for humidity in a non-linear way.* We apologize for the broken link, which has been removed. We carried out the validation and correction procedures but did not report them; we have now remedied this.

Validation against a reference instrument. All monitoring units were calibrated against a co-located E-BAM beta attenuation monitor, a US EPA Federal Equivalent Method reference instrument, before field deployment. Device-specific scatter plots of sensor readings against reference concentrations were generated; every unit achieved R^2 greater than 0.88 across the observed concentration range. We applied device-specific linear regression correction equations derived from these comparisons to all particulate matter data, so reported concentrations are expressed on the E-BAM scale. Units were re-checked against the reference instrument during the field period, with acceptable agreement defined *a priori* as a mean bias within $\pm 10\%$ of the reference concentration. We provide the device-specific scatter plots as Supplementary Figures 1 and 2. **Humidity.** We take the reviewer's point that optical sensors report mass that includes particle-bound water and that the relationship with relative humidity is not linear. Every device logs relative humidity and temperature concurrently, and we applied a correction for hygroscopic growth to all particulate matter data so reported concentrations represent dry particle mass. We now cite the relevant methodological literature for this correction.

Change made: New subsection "Instrument calibration and quality assurance" added under **Variables** (Methods); the inactive manufacturer link removed from "Ambient and indoor air quality monitor"; reference 61 added; Supplementary Figures 1 and 2 added.

Comment 4.4 *I am also a bit worried, as the other reviewer, about statistical power. If there are no big difference in exposure it might be even harder to find evidence.* This is the same concern Professor

Allen raised, and the reviewer is right that the driver is the exposure gradient rather than the sample size alone. We have therefore replaced the power statement entirely. Post-hoc power computed from an observed effect is a deterministic function of the p-value and conveys no additional information, so we no longer report it. Instead we report the effect sizes the study is designed to detect, against explicitly defined exposure contrasts. Assuming an LBW prevalence of 20% in the least-exposed group, two-sided α of 0.05, 80% power and an analysable sample of 369 women, the study can detect a risk ratio of approximately 1.9 (odds ratio approximately 2.5) for the highest versus lowest quartile of the pollutant distribution, and a risk ratio of approximately 1.6 (odds ratio approximately 2.0) for the above- versus below-median contrast. Treating exposure as continuous, the study can detect an odds ratio of approximately 1.44 per one standard deviation increase. If all 516 recruited women contribute outcome data, this becomes an odds ratio of approximately 2.2 for the extreme-quartile contrast and 1.36 per standard deviation. We now state plainly in the manuscript that, because the exposure contrast achievable within a single urban slum population is narrower than in studies spanning rural and urban settings, the study is powered to detect moderate rather than small associations. We will report the observed exposure distribution and interquartile range alongside every effect estimate so readers can judge the precision achieved. We think this is a more honest position than claiming high power.

Change made: The section "Sample size" has been renamed "**Sample size and detectable effect sizes**" and rewritten (Methods).

Comment 4.5 (implicit) *On the adequacy of two 24-hour measurements to represent exposure across pregnancy.* This concern is common to Professor Malmqvist's first comment and Professor Allen's review of version 2, and we had not previously answered it adequately. We now describe explicitly how we will use the two personal measurements. Each personal 24-hour measurement is treated as an individual-level anchor and is temporally adjusted against the continuous ambient record: we compute the ratio of the personal concentration to the co-temporal ambient concentration at the nearest State Pollution Control Board station, apply that participant-specific ratio to the ambient daily series across the second and third trimesters, and average the resulting series to obtain trimester-specific and pregnancy-average exposures. The assumption that the personal-to-ambient ratio is stable within an individual is testable with the two repeated measurements, and we will report the intraclass correlation coefficient between the second- and third-trimester measurements for each pollutant. Where the ICC is low, the primary analysis will be restricted to trimester-specific windows rather than pregnancy averages, and we will apply regression calibration to correct effect estimates for exposure measurement error. We state in the Discussion that the residual error is expected to be largely non-differential and to attenuate associations towards the null. **Change made:** New subsection "Deriving pregnancy-period exposure from repeated 24-hour measurements" added under **Variables** (Methods); corresponding limitation added to the **Discussion**.

Reviewer 5: Professor Chan Lu, Central South University, Changsha, China Comment 5.1 *In the abstract, the definition of "particulate matter (<10 μm , <2.5 μm)" is wrong. Should be "PM₁₀ (<=10 μm), PM_{2.5} (<=2.5 μm)".* Corrected. The notation is now PM₁₀ ($\leq 10 \mu\text{m}$) and PM_{2.5} ($\leq 2.5 \mu\text{m}$), applied consistently throughout the abstract, the main text, Table 1 and the list of abbreviations.

Change made: **Abstract** and throughout the manuscript.

Comment 5.2 *The structure is strange. In the introduction, "objectives: We aim to ..." This sentence should move to the abstract.* We apologise for the ambiguity in our previous reply, which was not clearly worded. The aims do appear in the Abstract, in the final sentence of the Background paragraph. The statement in the Introduction is a deliberate restatement, placed at the end of the section to

provide the transition into the Methods and to attach the two formal objectives to their hypotheses, which are not carried in the Abstract. We have restructured the end of the Background so that the two elements are clearly differentiated rather than reading as a repetition: the Background now closes with a short subsection headed “Objectives and hypotheses”, in which each numbered objective is immediately followed by its corresponding hypothesis. If the reviewer would still prefer the Introduction statement to be removed entirely, we will do so at the editor’s direction.

Change made: The closing part of the **Background** has been reorganised into a subsection headed “Objectives and hypotheses”.

Comment 5.3 *Figure 2 is strange. Please re-draw this map and make it more scientific.* Figure 2 has been redrawn. The revised figure presents the study setting in two labelled panels — (A) the state of Karnataka with Bengaluru Urban district highlighted, and (B) Bengaluru Urban district with the district boundary and the participating facilities — linked by a locator inset. Each panel now carries the cartographic elements that were missing from the previous version: a labelled graticule of latitude and longitude ticks, a north arrow, a segmented scale bar, a legend, a stated projection (Web Mercator, EPSG:3857) and base map attribution. The caption states that the scale bars are true at the central latitude of each panel.

Change made: **Figure 2** replaced; caption revised.

Comment 5.4 *The results is too simple. Please do more analysis especially including some sensitivity analysis stratified by personal factors.* This manuscript is submitted to Wellcome Open Research as a Study Protocol. Under that article type, the manuscript sets out the design, methods, and planned procedures of the APPLE cohort; it does not report results, so it does not include a results section. The primary results papers arising from the cohort will report empirical findings, where the analyses the reviewer describes will be presented in full. We agree entirely that stratified analysis is essential, and we have specified it prospectively rather than leaving it to be decided after the data are seen. The planned analysis now includes sensitivity analyses stratified by maternal age, socio-economic status, parity, occupational history of exposure, anaemia status and primary cooking fuel, together with sensitivity analyses examining trimester-specific versus pregnancy-average exposure windows, regression calibration for measurement error, the influence of the humidity correction, and departures from the live-birth-bias assumption.

Change made: New paragraph “Planned sensitivity and stratified analyses” added under **Statistical methods** (Methods).

Comment 5.5 *The discussion section is too short. This is a research article, which should include a relatively long and rich discussion comparing your results with other studies.* As above, the submission is a Study Protocol rather than an original research article, and in line with reporting guidance for protocols, the discussion is directed at the rationale, feasibility and anticipated impact of the design rather than at the interpretation of findings. A discussion comparing results with the published literature would require results, which this article does not contain. We have made the article type explicit in the Discussion so that this is clear to readers. We accept, however, that the previous Discussion was thin even by protocol standards. It has been expanded to include a Strengths paragraph, a Limitations paragraph covering exposure measurement error, the narrow exposure gradient and live-birth bias, and a paragraph on environmental justice and future work.

Change made: **Discussion** expanded with three new paragraphs and an explicit statement of article type.

Comment 5.6 *I suggest the authors adding some important references in the introduction and/or discussion: Lu C et al. 2020; Lu C et al. 2019; Lu C et al. 2021.* We thank the reviewer for these suggestions, which are directly relevant to the design of this study. Rather than adding them as an

undifferentiated block, we have integrated them where they carry argumentative weight: they are the strongest available demonstration that ambient and traffic-related pollution act jointly with indoor environmental factors on birth outcomes, which is precisely the rationale for measuring indoor and ambient concentrations concurrently, as APPLE does. They are cited in a new paragraph in the Background making that argument, and again in the sentence listing environmental determinants of intrauterine growth restriction. **Change made:** References 53 (Chemosphere 2020), 54 (Ecotoxicol Environ Saf 2019) and 55 (Build Environ 2021) added and cited in the **Background**.

Reviewer 6: Dr Jiawen Liao, University of Southern California, USA Comment 6.1 *Figure 2 showing the map looks like they are scaled. I would recommend authors to provide better quality maps, preferably using ArcGIS or QGIS software.* Figure 2 has been replaced. The revised figure is correctly scaled and projected, states its projection explicitly (Web Mercator, EPSG:3857), and carries a labelled coordinate graticule on both panels so that the extent and scaling can be verified directly from the figure. We added a north arrow, segmented scale bar, and legend to each panel, and a locator inset linking the state and district panels. We have also corrected a spelling error in one of the facility labels.

Change made: Figure 2 replaced.

Comment 6.2 *The statement about the skinfold thickness equation being correlated and validated against DEXA needs a reference.* References have been added. We now cite the validation of skinfold-thickness prediction equations against DEXA in South Asian children, in which measurements were made with Holtain callipers and the Slaughter equation showed reasonable validation against DEXA with high accuracy, minimal bias and good precision. We also cite the corresponding work relating measured and derived anthropometry to DEXA-measured fat and lean mass in neonates, since our earliest measurements are made at birth. The sentence has been rewritten so that the claim made is the one the cited evidence supports, rather than a general claim of validation.

Change made: References 59 and 60 added; the **Adiposity** paragraph rewritten.

Comment 6.3 *Blood sample collection: from the protocol, it looks like authors are just collecting fasting blood samples from pregnant women in a hospital. Authors shouldn't mention that "assays will be carried out" in the manuscript, because the activity is only involved sample collection.* We take the reviewer's point that the previous wording did not make clear who performs which activity, and we have rewritten the section to state the division of responsibility explicitly rather than leaving it implied. The assays are part of this study and are performed for it. The research institution holds a memorandum of understanding with Medall Healthcare Pvt. Ltd (MEDALL), a nationally accredited central laboratory, under which the laboratory investigations planned in this protocol — haemoglobin and the 2-hour OGTT (fasting and postprandial) — are carried out. The research team is responsible for participant preparation, venepuncture, on-site processing, aliquoting and storage; MEDALL performs the assays. The revised text states this directly, so that no reader is left to infer that the investigators are running the assays themselves, and equally so that the assays are not mistaken for activity outside the study.

Change made: Opening paragraph of **Laboratory analysis and sample storage** rewritten.

Comment 6.4 *Blood sample collection time: authors should provide some more details at what point of pregnancy the blood is collected, at first, second or third trimester?* The timing is now stated explicitly. Blood is collected once, at or after 24 weeks of gestation — that is, in the late second or early third trimester. No blood samples are collected in the first trimester. This is consistent with Table 1, which gives the window as 24–36 weeks of gestation.

Change made: Second paragraph of **Laboratory analysis and sample storage**.

Reviewer 7: Dr Francisco Jablinski Castelhana, Federal University of Rio Grande do Norte, Brazil

Comment 7.1 *There's a clear relationship between air pollution and socioeconomical factors in India ... so this is a clear case of environmental justice. Despite the fact that you studied only the slums of Bangalore, perhaps a comparison with other parts of the city that are not slums could be interesting.* We agree that this is an environmental justice problem, and the reviewer has identified a real boundary of what this cohort can show. Our design decision was deliberate: the objective is to characterise exposure and risk within the population that carries the heaviest burden and is least represented in the existing evidence, and recruiting a non-slum comparison arm within the same budget would have reduced the precision available in the slum population itself. We accept that this means the study cannot quantify the slum versus non-slum disparity, and we now say so rather than leaving it unstated. A new paragraph in the Discussion frames the design choice, states the limitation explicitly, and identifies a slum versus adjacent non-slum comparison in Bengaluru as a priority for future work, extending to peri-urban and rural settings.

Change made: New paragraph "Environmental justice and future work" added to the **Discussion**.

Comment 7.2 *I also believe that the adiposity is not quite well explained. Recommend that you take more time to explain how air pollution could mediate the adiposity situation on the babies.* A dedicated paragraph has been added to the Background setting out the mechanistic chain. In summary: maternal exposure to particulate pollutants induces placental dysfunction and low-grade systemic inflammation, which disrupt fetal energy homeostasis and adipokine regulation; cohort studies report altered cord-blood leptin and adiponectin concentrations and greater infant weight and adiposity following prenatal exposure to traffic-related and ambient particulate pollution; these signals impair insulin signalling and promote lipidomic changes that appear to programme altered adipose tissue development; and fetal growth restriction arising through these pathways is characteristically followed by accelerated postnatal catch-up growth that preferentially deposits central rather than peripheral fat, so that infants born small accumulate disproportionate adipose tissue relative to lean mass. This gives the biological rationale for treating low birth weight as a mediator in Hypothesis 2, which the previous version asserted without explaining.

Change made: New paragraph added to the **Background**; references 56–59 added. **Comment 7.3** *I strongly recommend to remake the map of figure 2 to make it more scientific by using any GIS software, by doing this, you should add cartographical elements such as scale, wind rose and others.* We agree, and we have added each of the elements the reviewer names. The revised Figure 2 carries, on both panels: a segmented scale bar in kilometres, true at the central latitude of the panel; a north arrow; a graticule with labelled latitude and longitude ticks around the frame; a legend; an explicit statement of the projection (Web Mercator, EPSG:3857); and base map attribution. The two panels are arranged as a state-to-district hierarchy with a locator inset. Because the panels are in Web Mercator, the scale varies slightly with latitude across the state-level panel, and we state in the caption that the scale bar is true at the central latitude rather than implying a single constant scale.

Change made: **Figure 2** replaced; caption revised.

Comment 7.4 *Besides that, I recommend the reading of the following paper that could also help you with the LBW and air pollution matter.* We would very much like to incorporate this reference, but the citation itself does not appear in the review report as transmitted to us; the sentence ends without the reference being given. We would be grateful if the reviewer could supply it, and we will cite it in the next version. In the meantime we have strengthened the coverage of the air pollution and low

birth weight literature in the Background, including the three references suggested by Professor Lu and additional work on prenatal exposure and infant adiposity.

Change made: References 53–59 added and cited in the **Background**. **Points carried over from earlier reviews** Dr Liao notes that he read the reports of Professor Balakrishnan, Professor Allen and Professor Marshall. Two of the concerns raised in those reports were not adequately resolved in version 3, and we address them directly here.

Professor Allen (version 2) *The investigators plan to use the VAYUCARE device for measuring air pollutants, but the paper does not provide sufficient detail on this instrument or a rationale for choosing it. Has the device been validated against other, more established monitors? ... I agree with Dr. Marshall's recent comments that the VAYUCARE needs to be compared to a "gold standard" device and that six hours is insufficient for such a comparison.* This is now addressed in the new "Instrument calibration and quality assurance" subsection. Every unit was compared against a co-located E-BAM Federal Equivalent Method reference instrument, achieving R^2 greater than 0.88, and device-specific linear correction equations were derived and applied. Device-specific scatter plots are provided as supplementary material so that reviewers and readers can assess the agreement themselves rather than relying on our summary of it. Units were re-checked against the reference during the field period with an *a priori* acceptance criterion of $\pm 10\%$ mean bias.

Professor Allen (version 2) *It remains unclear how two 24-hour measurements will be used to estimate exposure over the full pregnancy or during specific trimesters. These short-term measurements will be heavily influenced by temporal variations in concentrations, and that variation will need to be accounted for in some way.*

This is now addressed in the new subsection "Deriving pregnancy-period exposure from repeated 24-hour measurements", described under Comment 4.5 above. In brief, personal measurements are used as individual-level anchors, scaled to the continuous ambient series by a participant-specific personal-to-ambient ratio, and averaged over trimester-specific and pregnancy-length windows. The stability assumption underlying that scaling is tested empirically using the two repeated measurements via the ICC, with a pre-specified fallback to trimester-specific windows and regression calibration where the ICC is low. We also state the expected direction of residual bias.

Professor Allen (version 2) *A key driver of statistical power is the exposure gradient, but the power calculation still does not specify what gradient is being assumed.*

The power section has been rewritten to report detectable effect sizes against explicitly defined exposure contrasts — extreme quartiles, median split, and a one standard deviation increase — rather than a single power figure. We now state that the study is powered to detect moderate rather than small associations because the exposure gradient within a single urban slum population is narrower than in rural-urban designs, and we commit to reporting the observed exposure distribution alongside every effect estimate.

Professor Allen (version 2) *Given emerging evidence that air pollution contributes to pregnancy loss, I encourage the authors to think through the potential role of "live birth bias" in this study.*

The Bias section now names live-birth bias explicitly, cites the literature the reviewer identified, and commits to recording reasons for pregnancy loss and neonatal death and to examining the sensitivity of findings to plausible selection scenarios. The limitation is also stated in the Discussion.

Change made: Bias section revised; references 62 and 63 added; limitation added to the **Discussion**. We hope that these revisions address the reviewers' concerns. We are grateful for the depth of the review this protocol has received, and we are happy to make further changes should any point remain unresolved.

Competing Interests: No competing interests were disclosed.
